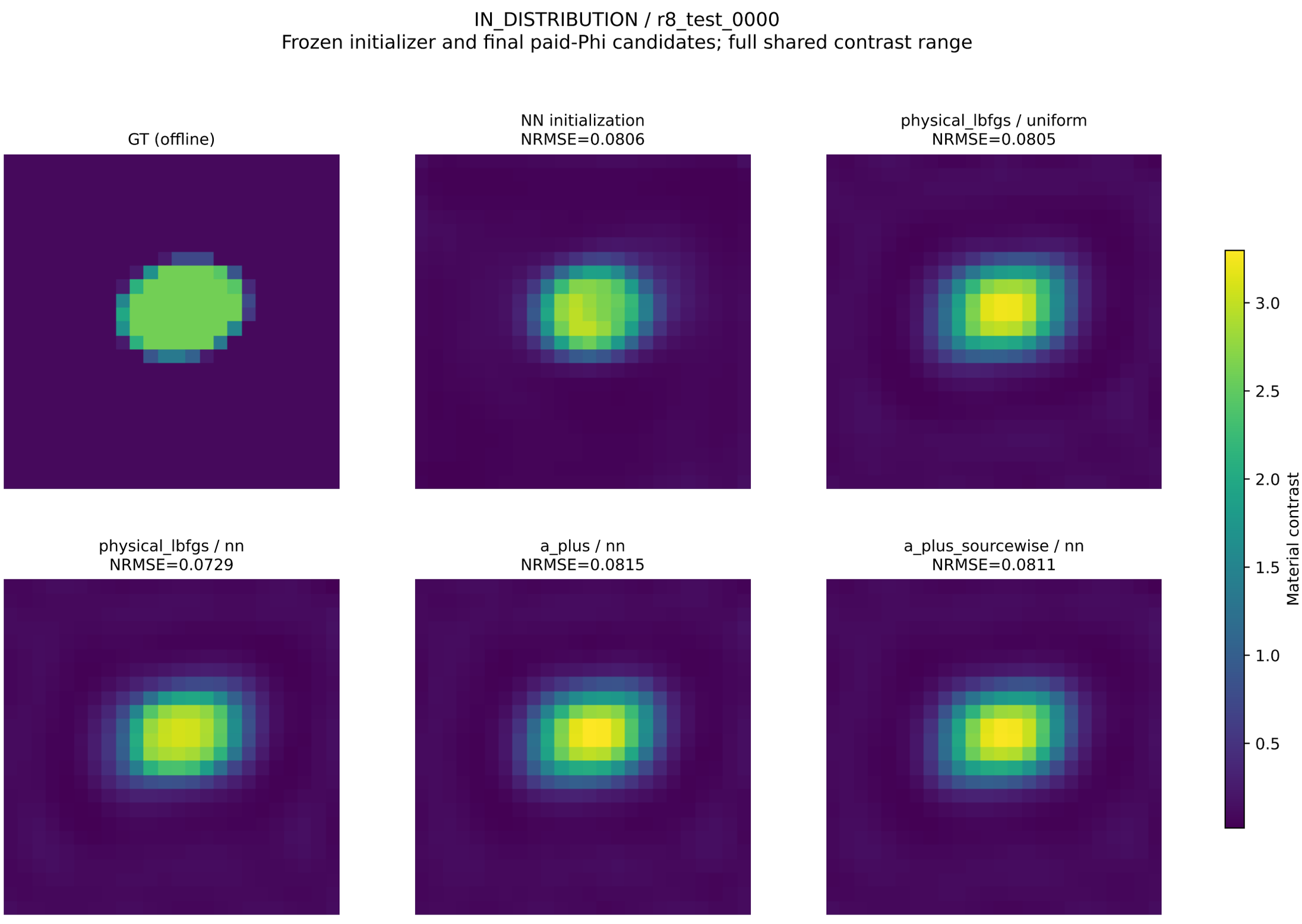
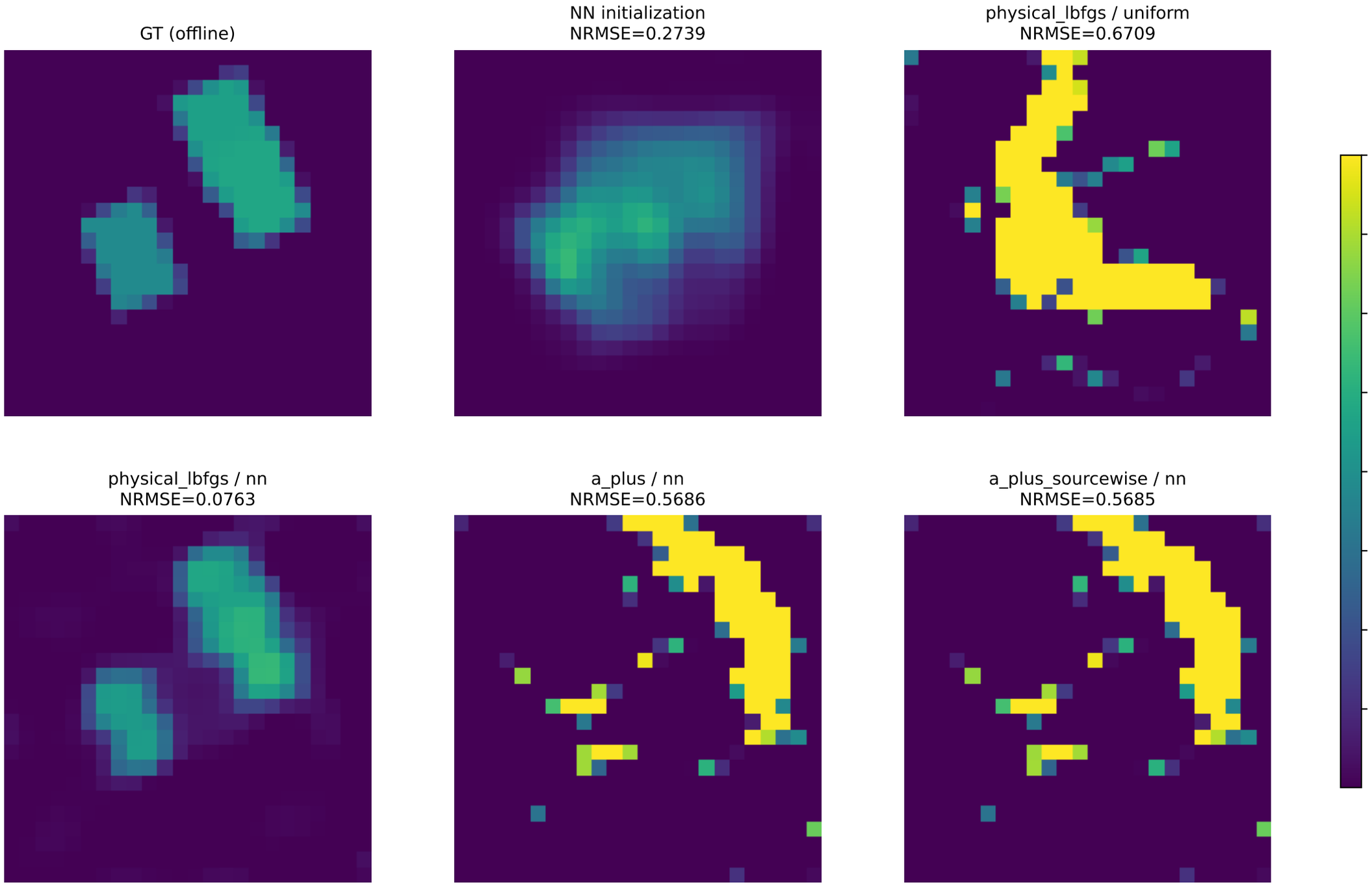
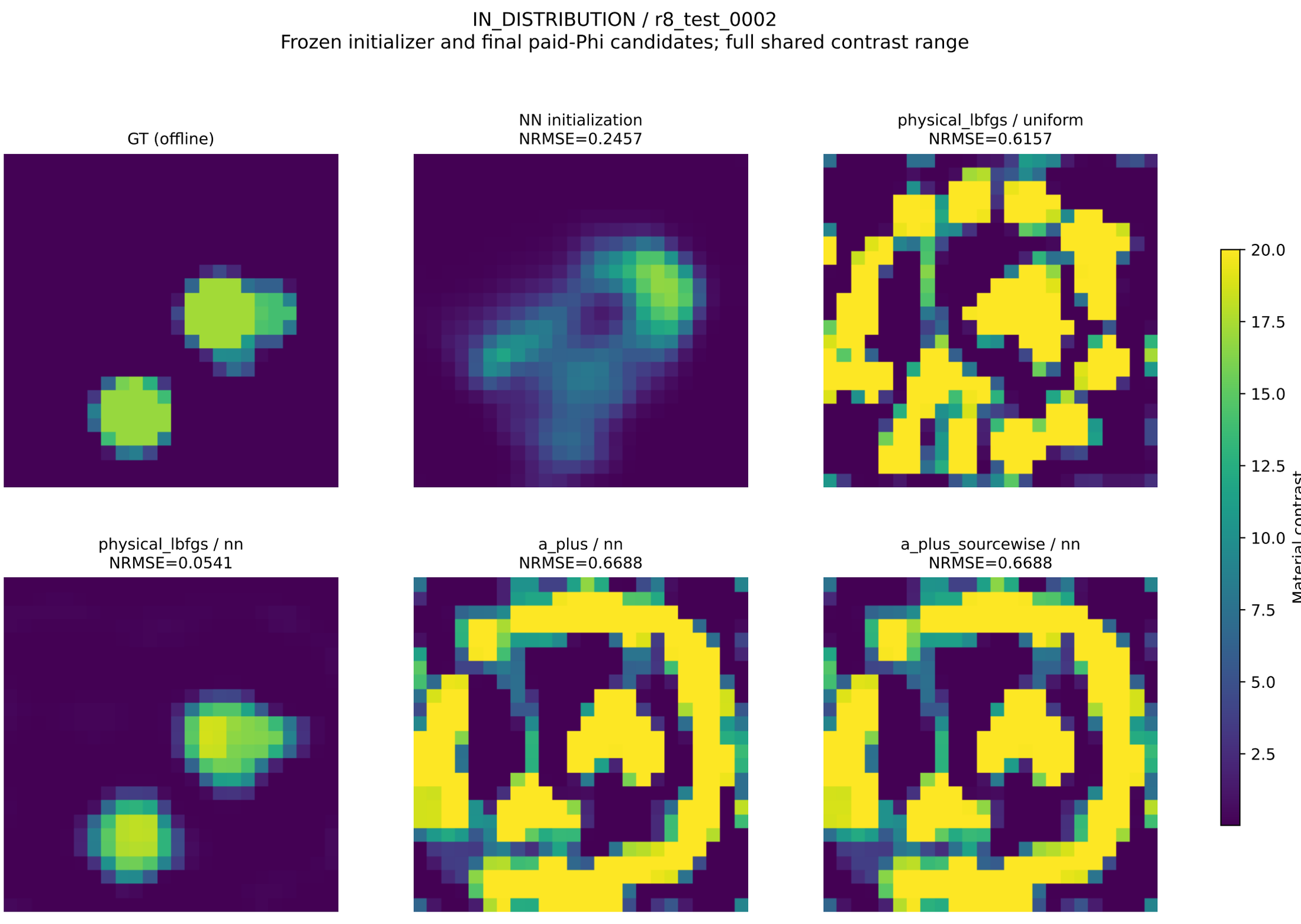
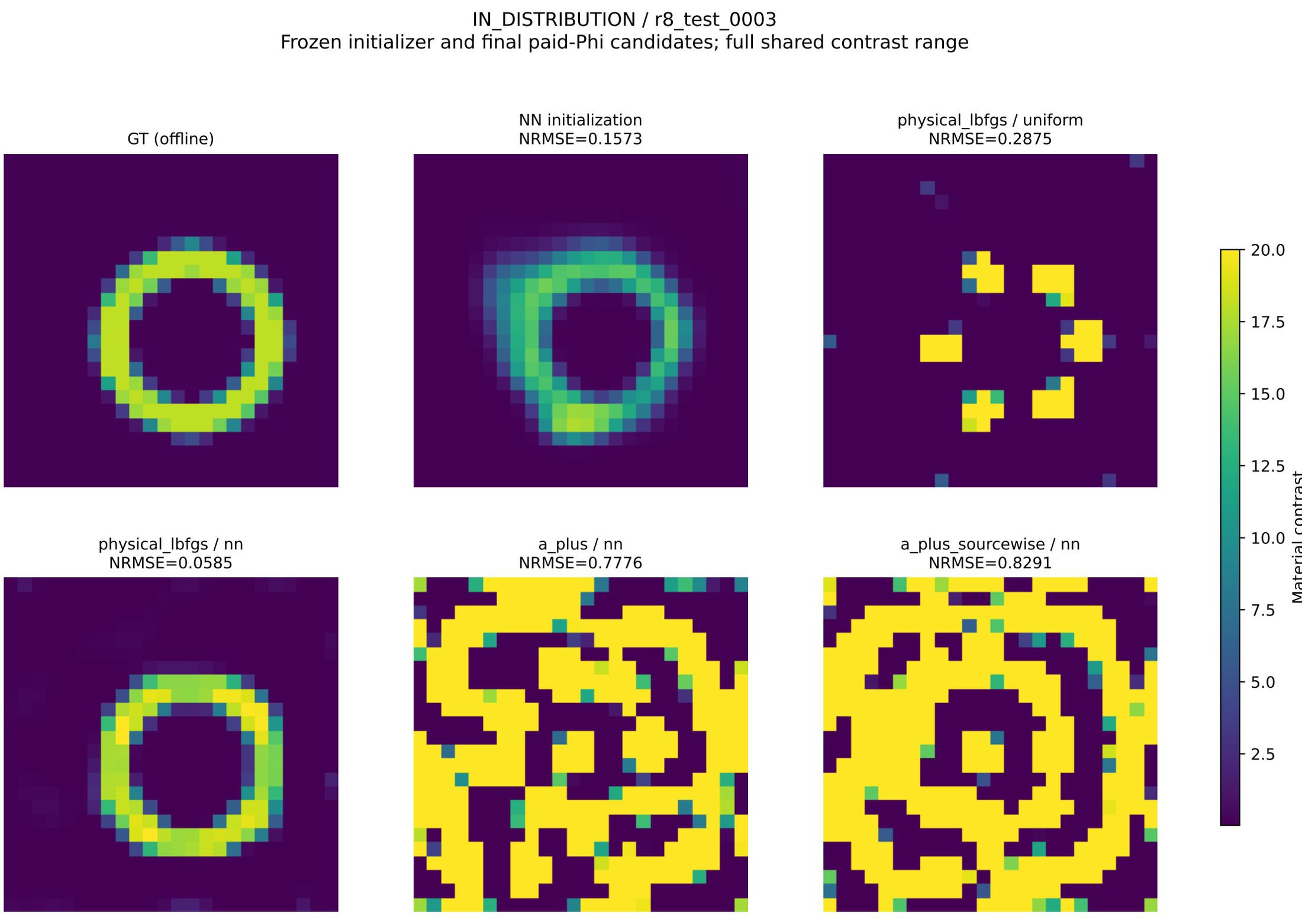
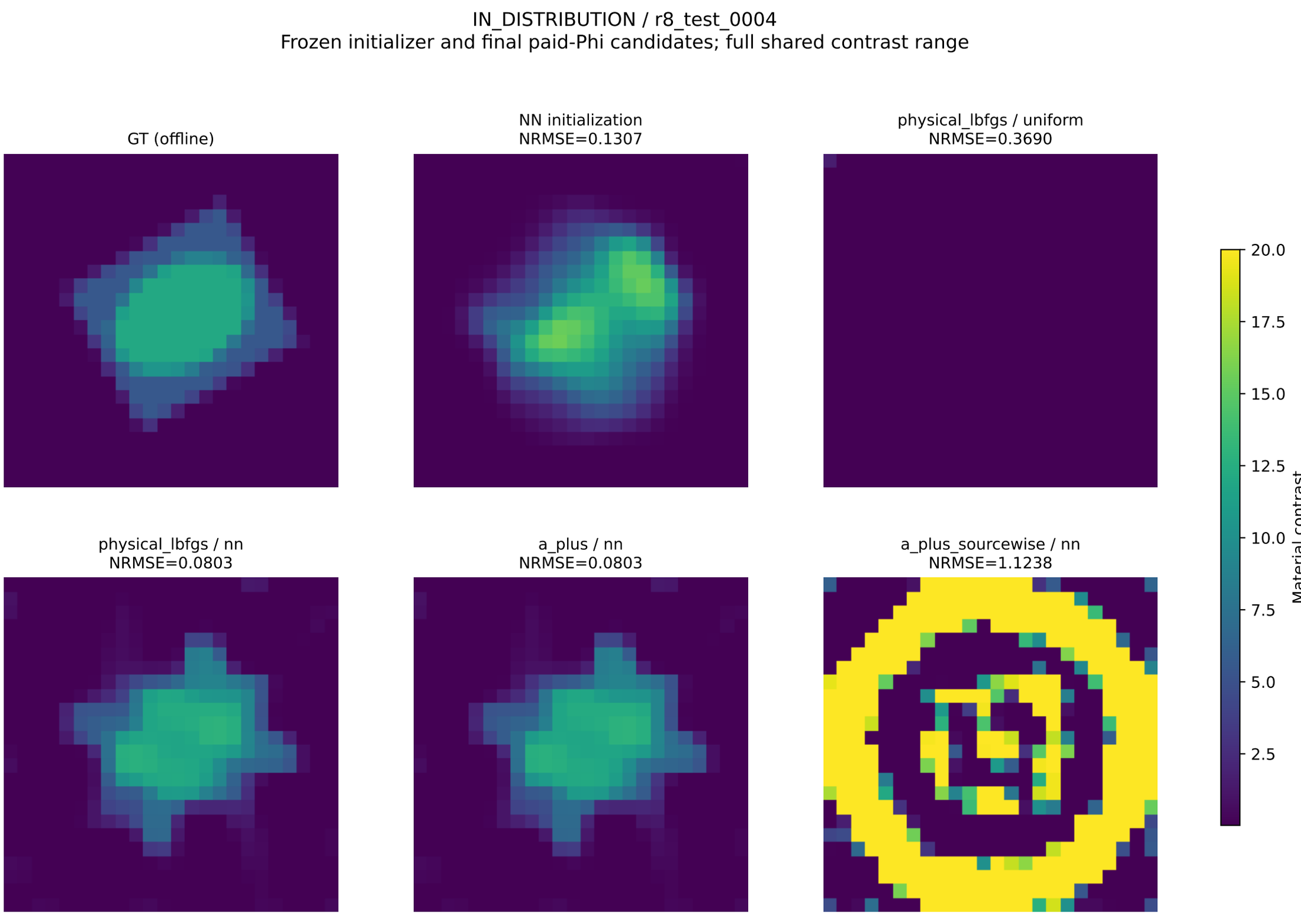
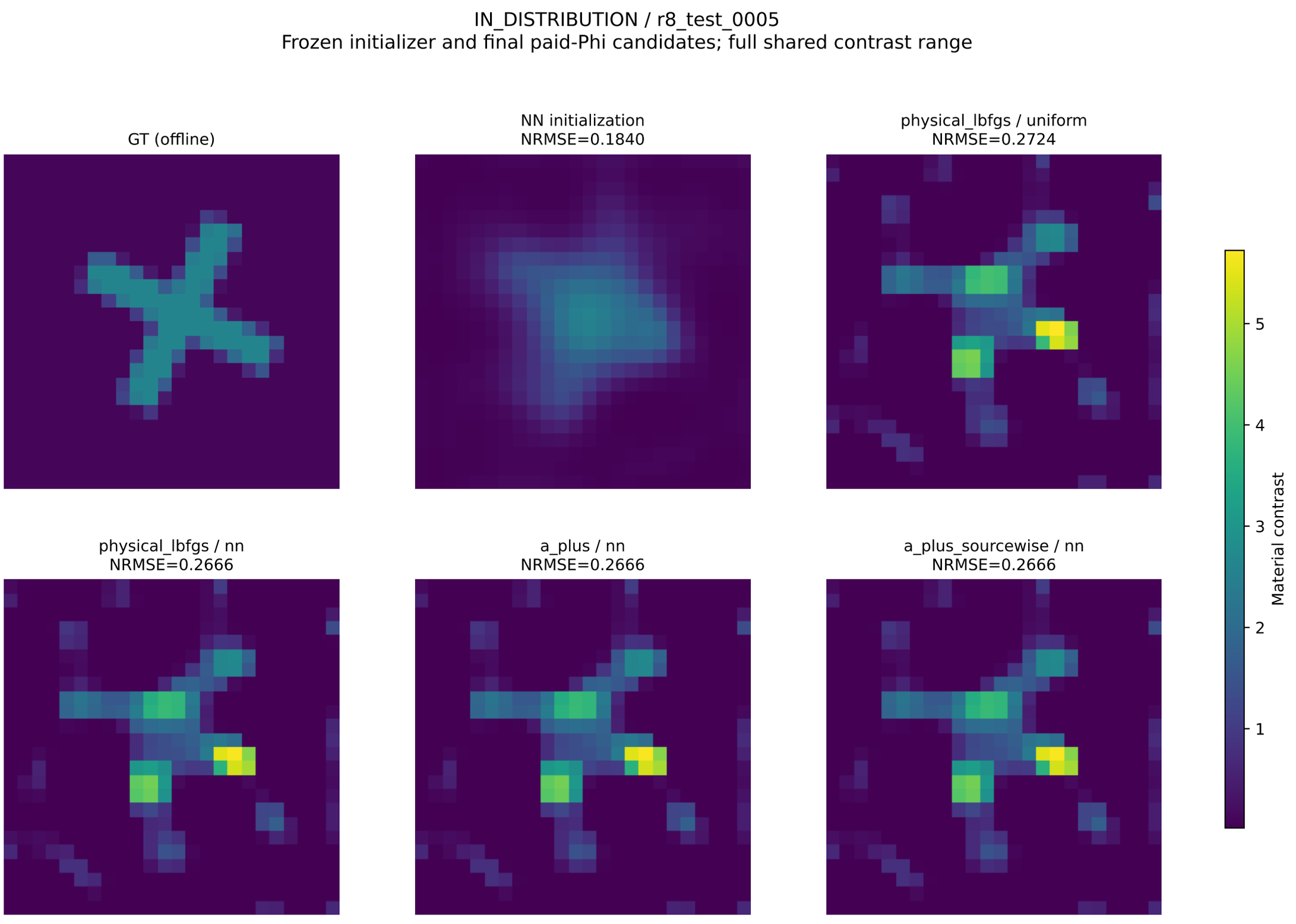

IN_DISTRIBUTION / r8_test_0000
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0806
physical_lbfgs / uniform
NRMSE=0.0805
physical_lbfgs / nn
NRMSE=0.0729
a_plus / nn
NRMSE=0.0815
a_plus_sourcewise / nn
NRMSE=0.0811
Material contrast
3.0
2.5
2.0
1.5
1.0
0.5

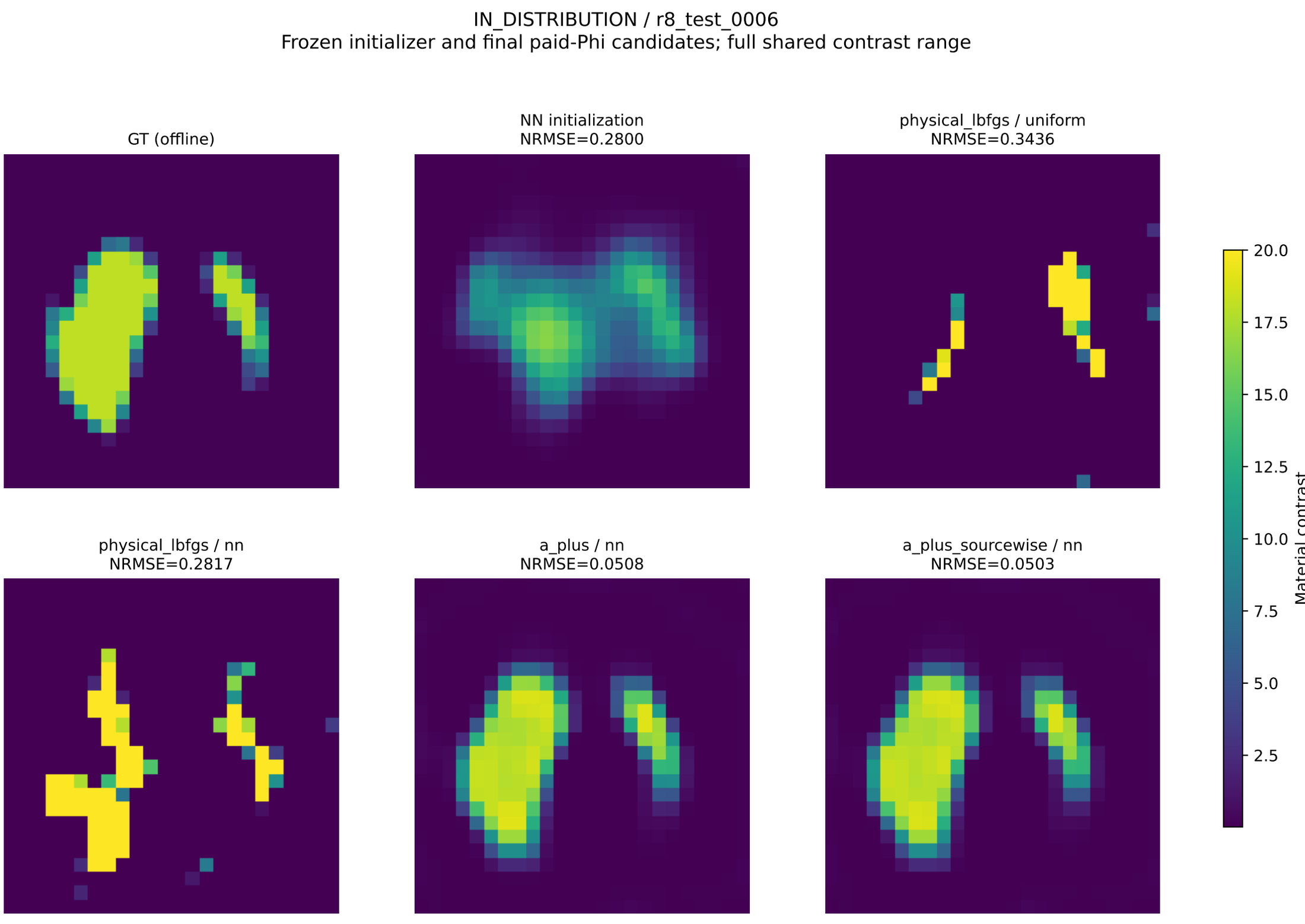

IN_DISTRIBUTION / r8_test_0001
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2739
physical_lbfgs / uniform
NRMSE=0.6709
physical_lbfgs / nn
NRMSE=0.0763
a_plus / nn
NRMSE=0.5686
a_plus_sourcewise / nn
NRMSE=0.5685
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

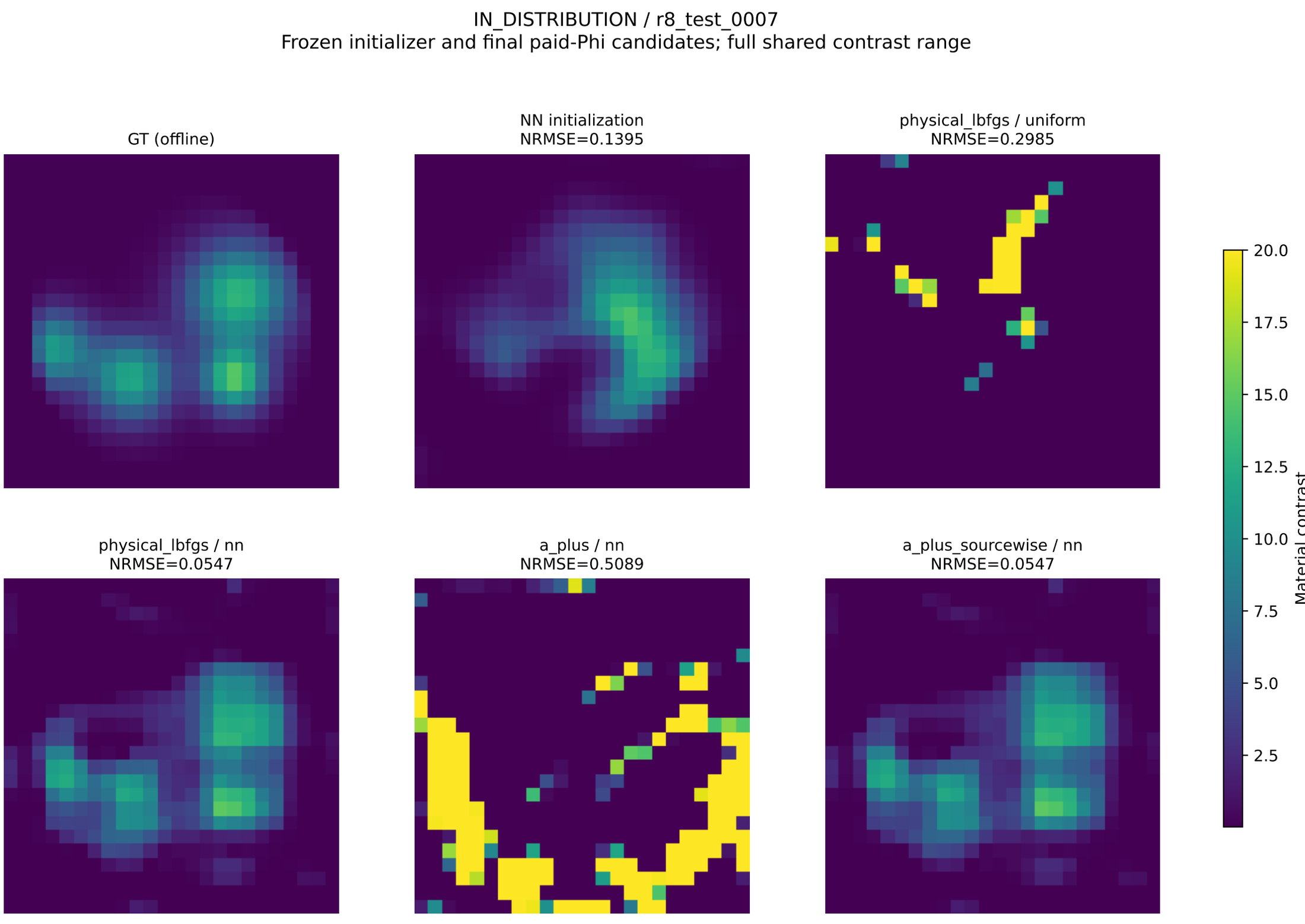

IN_DISTRIBUTION / r8_test_0002
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2457
physical_lbfgs / uniform
NRMSE=0.6157
physical_lbfgs / nn
NRMSE=0.0541
a_plus / nn
NRMSE=0.6688
a_plus_sourcewise / nn
NRMSE=0.6688
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

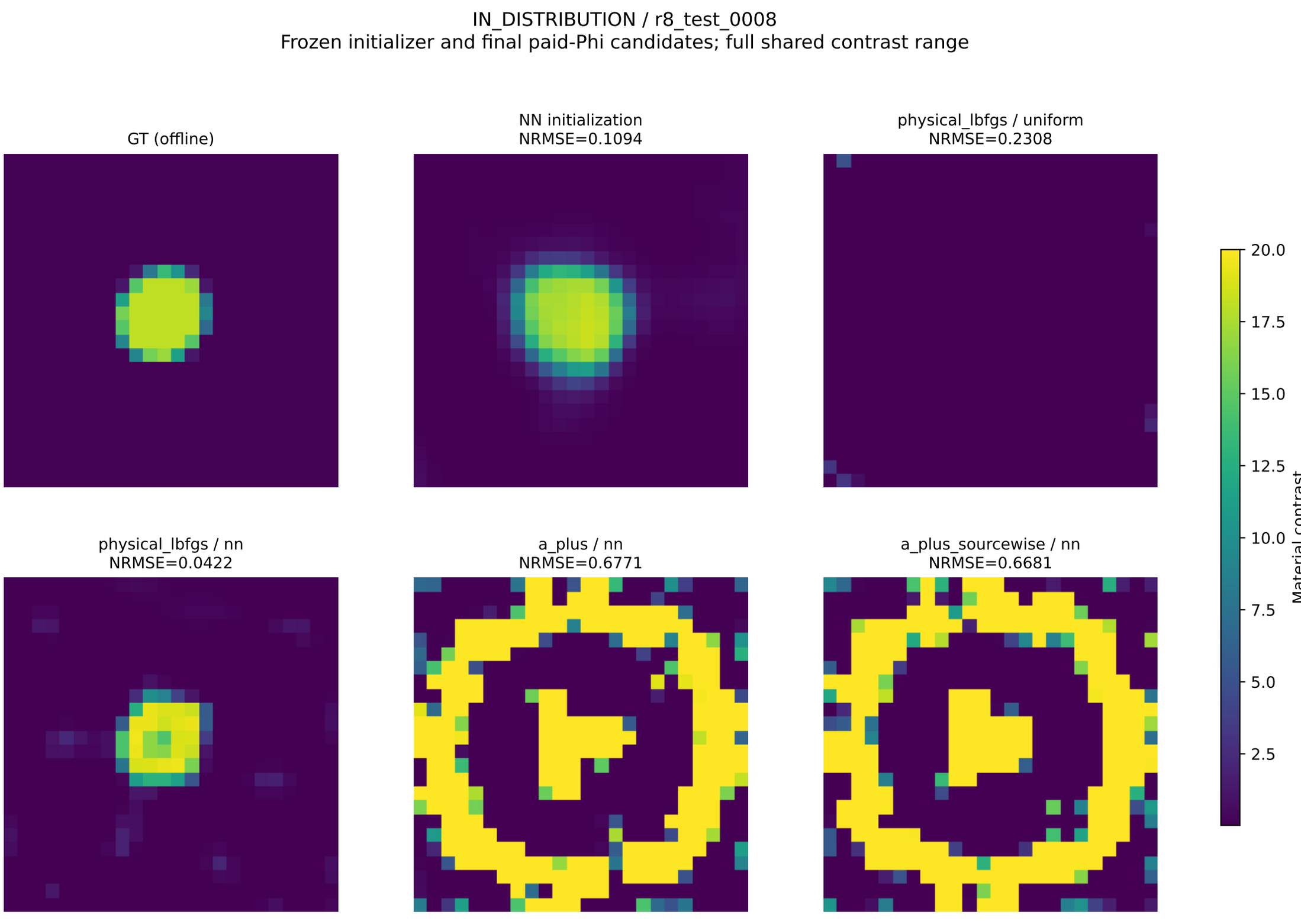

IN_DISTRIBUTION / r8_test_0003
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1573
physical_lbfgs / uniform
NRMSE=0.2875
physical_lbfgs / nn
NRMSE=0.0585
a_plus / nn
NRMSE=0.7776
a_plus_sourcewise / nn
NRMSE=0.8291
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

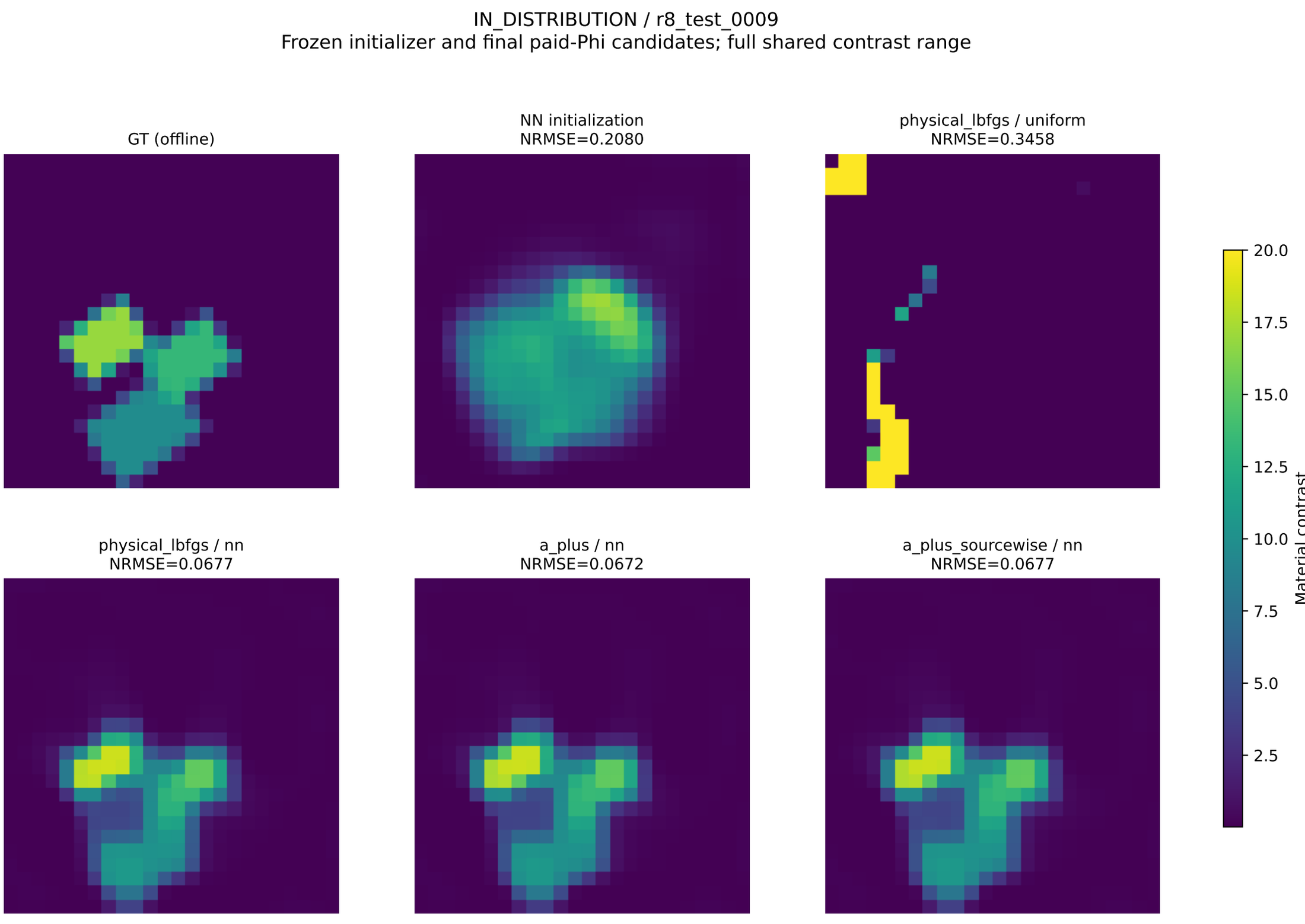

IN_DISTRIBUTION / r8_test_0004
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1307
physical_lbfgs / uniform
NRMSE=0.3690
physical_lbfgs / nn
NRMSE=0.0803
a_plus / nn
NRMSE=0.0803
a_plus_sourcewise / nn
NRMSE=1.1238
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

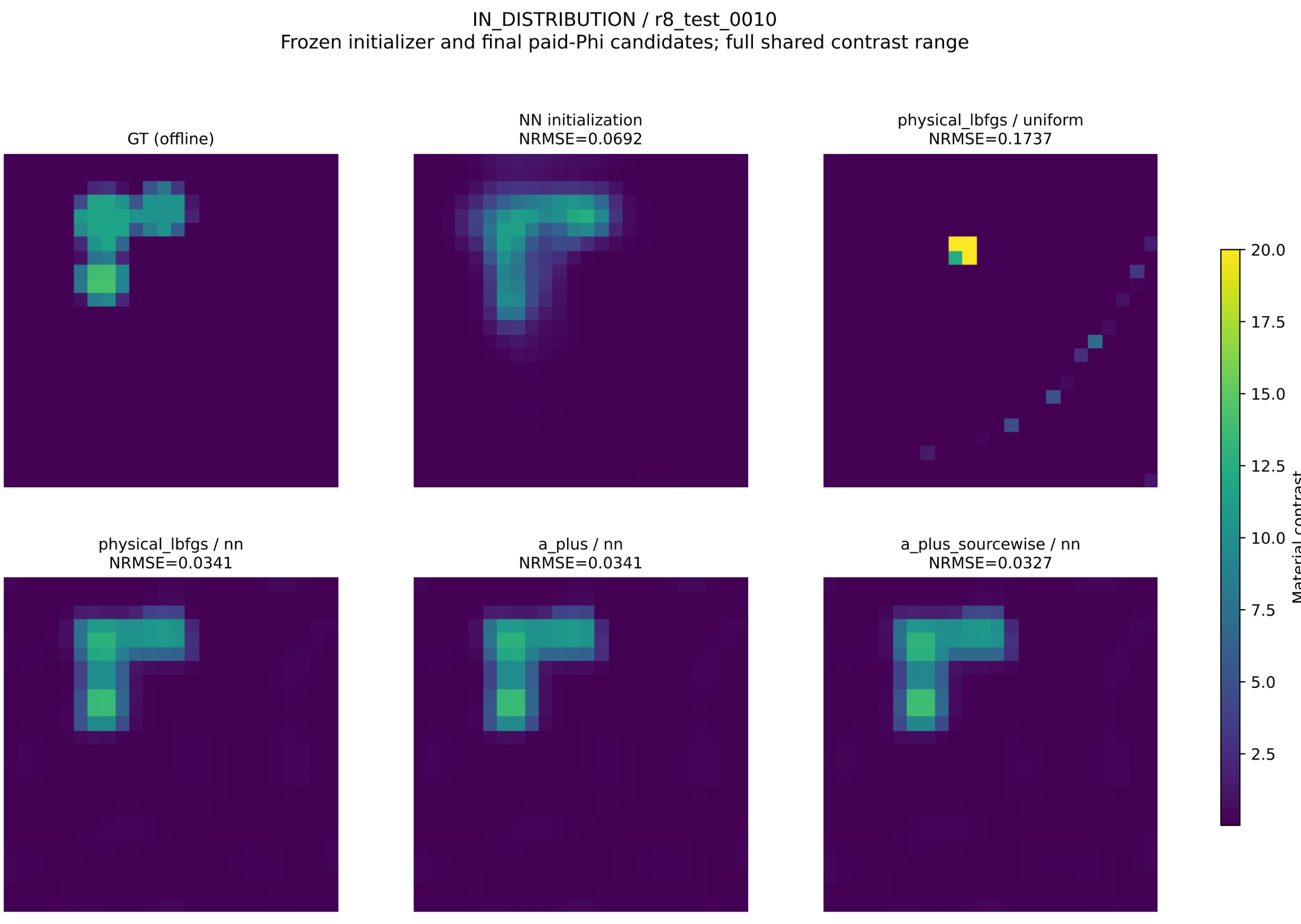

IN_DISTRIBUTION / r8_test_0005
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1840
physical_lbfgs / uniform
NRMSE=0.2724
physical_lbfgs / nn
NRMSE=0.2666
a_plus / nn
NRMSE=0.2666
a_plus_sourcewise / nn
NRMSE=0.2666
Material contrast
1
2
3
4
5

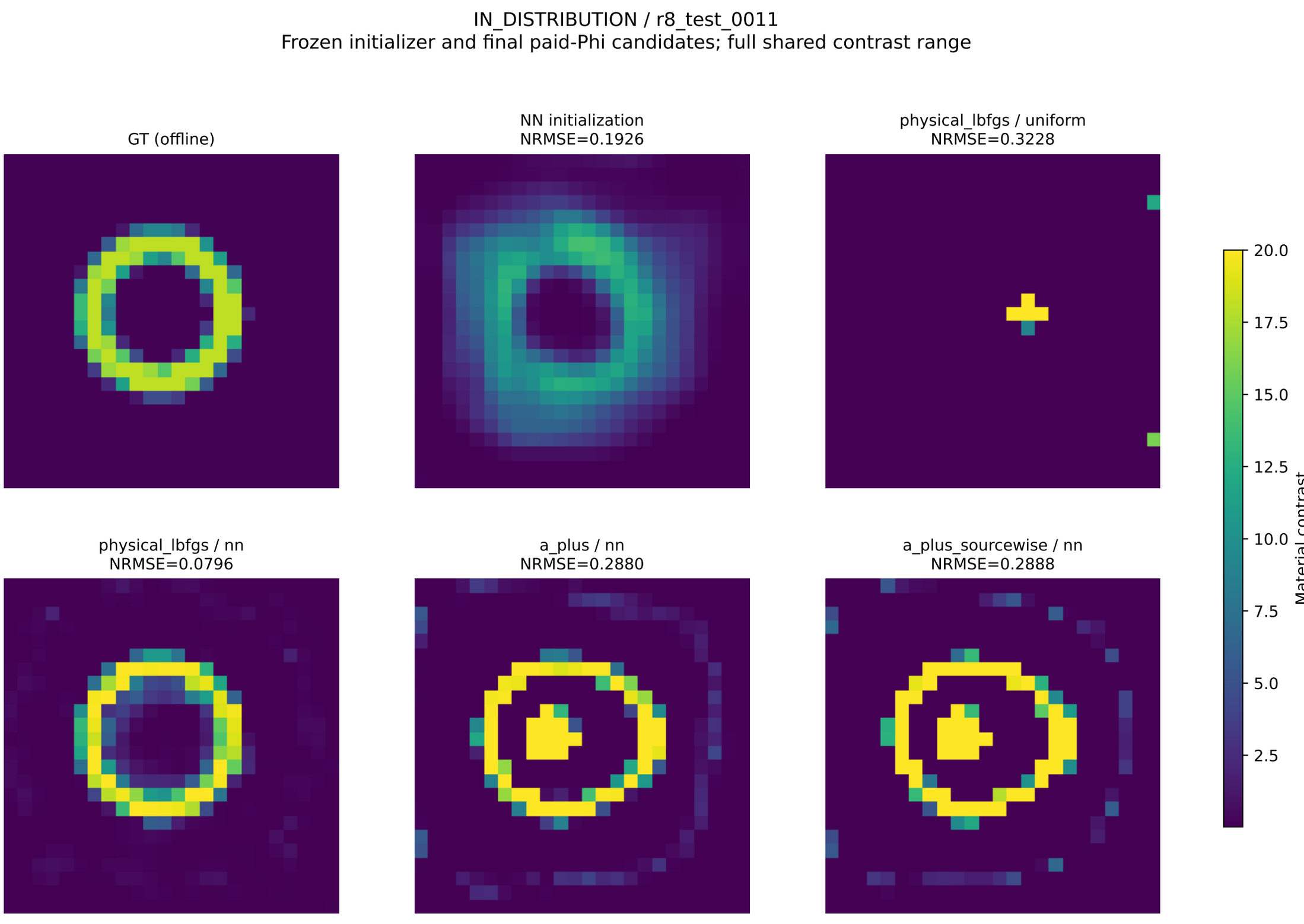

IN_DISTRIBUTION / r8_test_0006
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2800
physical_lbfgs / uniform
NRMSE=0.3436
physical_lbfgs / nn
NRMSE=0.2817
a_plus / nn
NRMSE=0.0508
a_plus_sourcewise / nn
NRMSE=0.0503
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

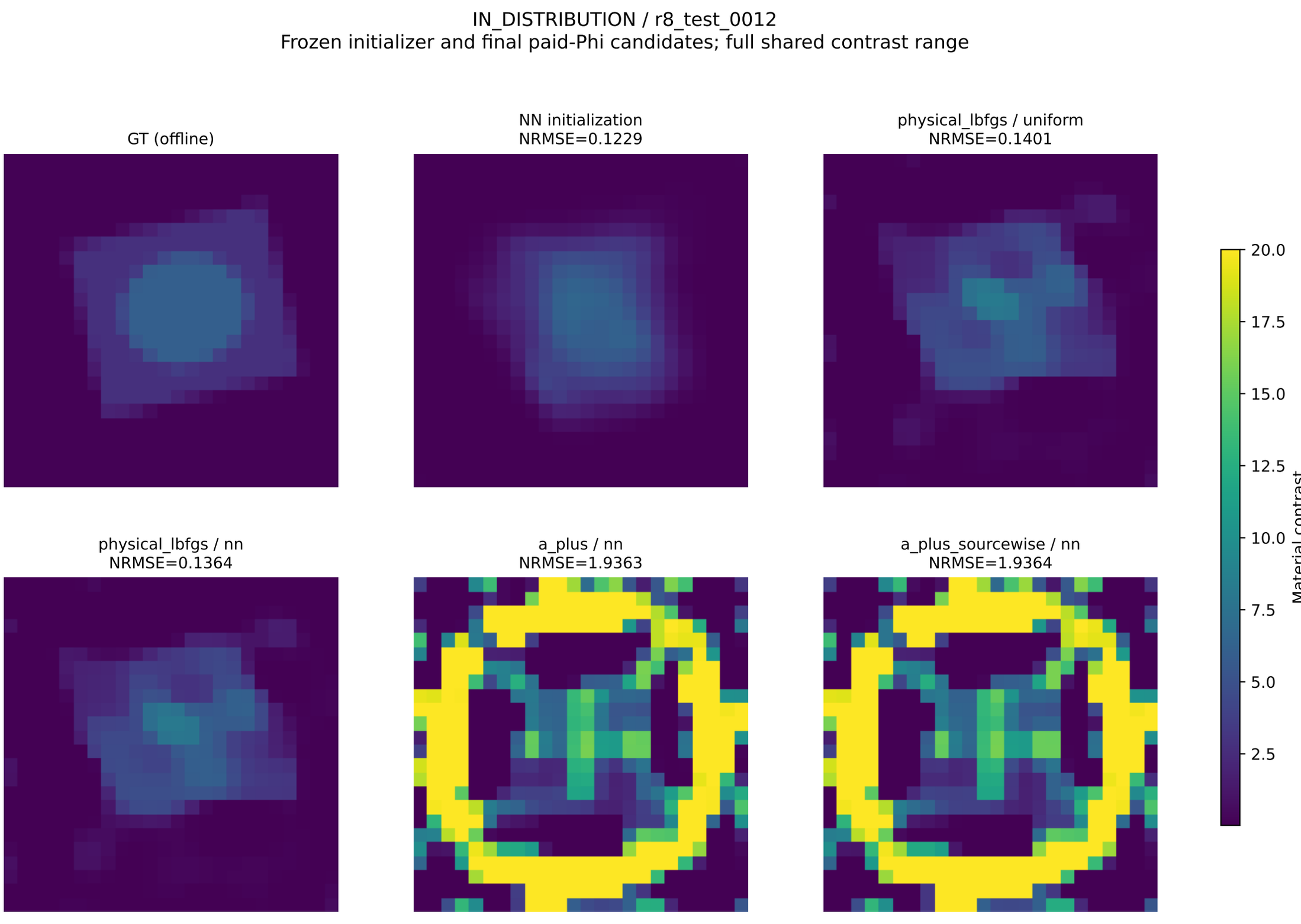

IN_DISTRIBUTION / r8_test_0007
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1395
physical_lbfgs / uniform
NRMSE=0.2985
physical_lbfgs / nn
NRMSE=0.0547
a_plus / nn
NRMSE=0.5089
a_plus_sourcewise / nn
NRMSE=0.0547
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

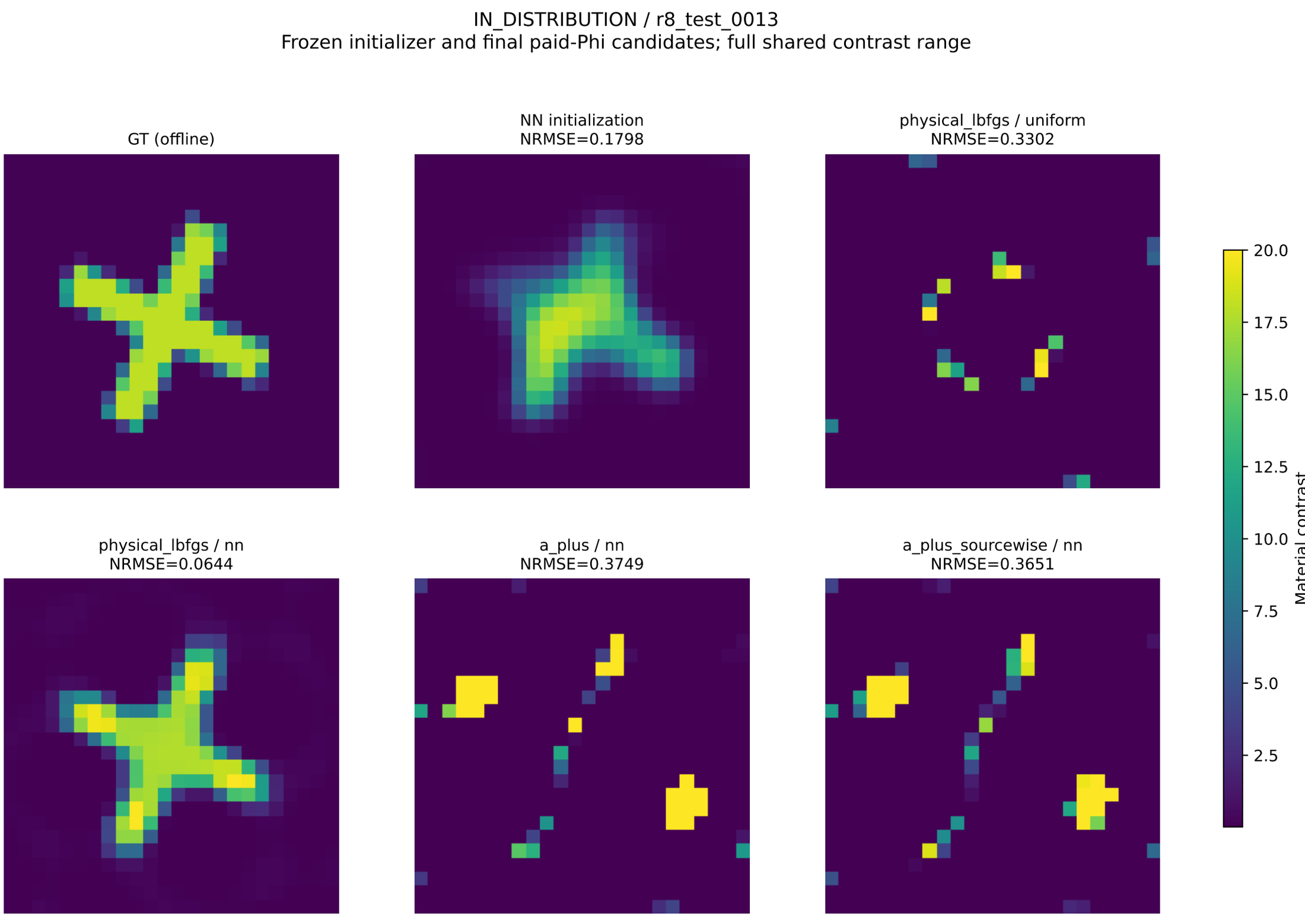

IN_DISTRIBUTION / r8_test_0008
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1094
physical_lbfgs / uniform
NRMSE=0.2308
physical_lbfgs / nn
NRMSE=0.0422
a_plus / nn
NRMSE=0.6771
a_plus_sourcewise / nn
NRMSE=0.6681
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

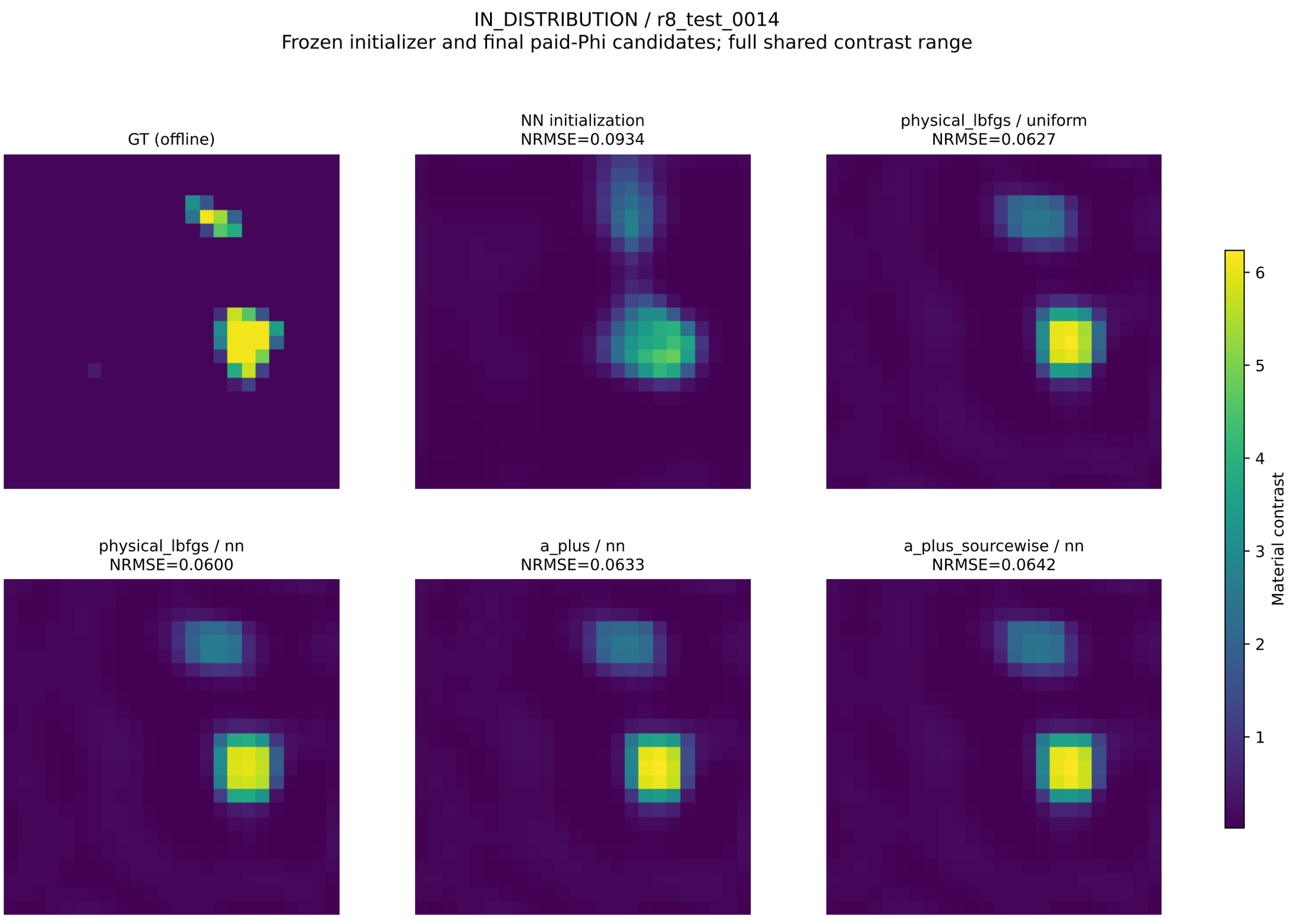

IN_DISTRIBUTION / r8_test_0009
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2080
physical_lbfgs / uniform
NRMSE=0.3458
physical_lbfgs / nn
NRMSE=0.0677
a_plus / nn
NRMSE=0.0672
a_plus_sourcewise / nn
NRMSE=0.0677
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

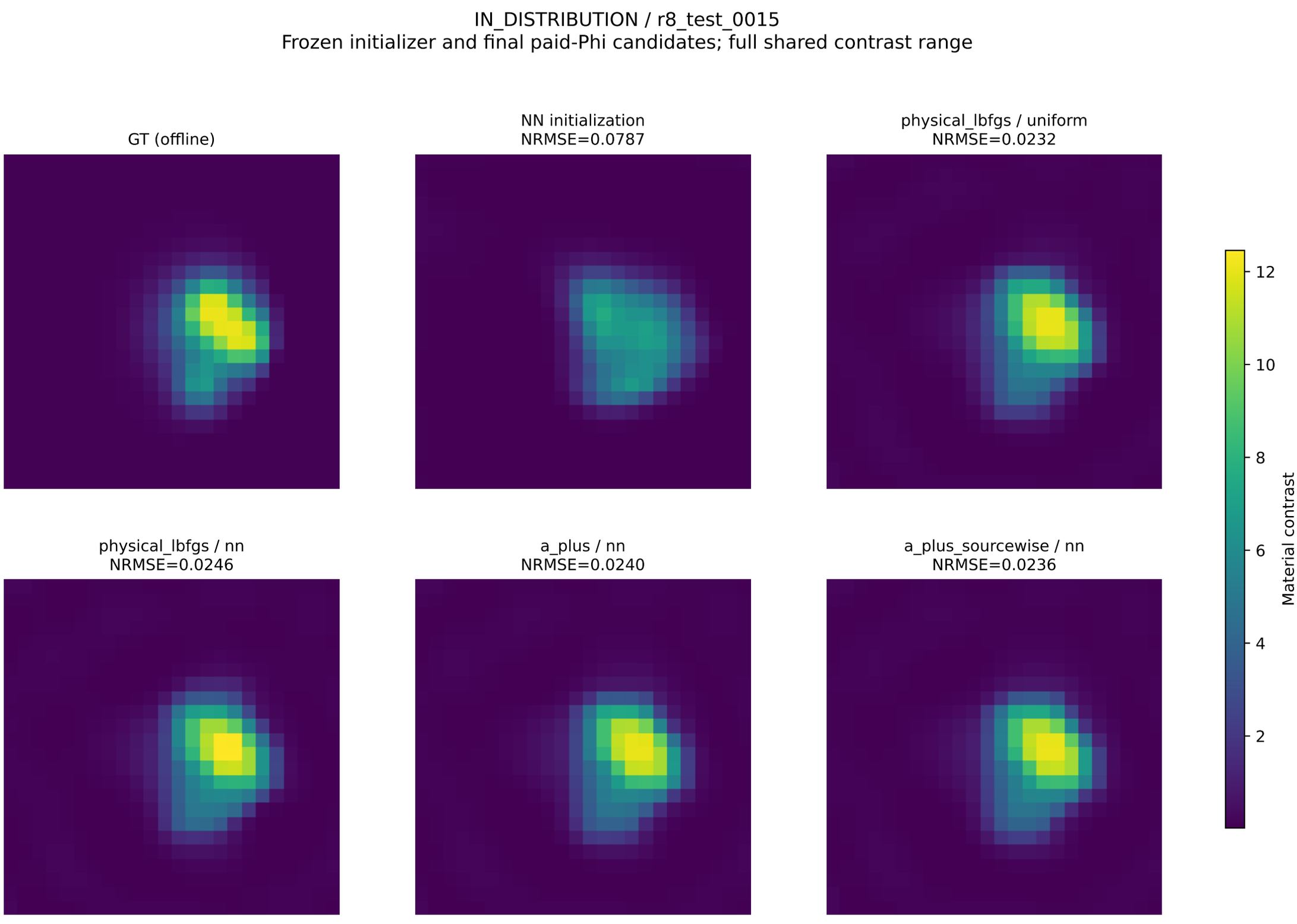

IN_DISTRIBUTION / r8_test_0010
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0692
physical_lbfgs / uniform
NRMSE=0.1737
physical_lbfgs / nn
NRMSE=0.0341
a_plus / nn
NRMSE=0.0341
a_plus_sourcewise / nn
NRMSE=0.0327
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

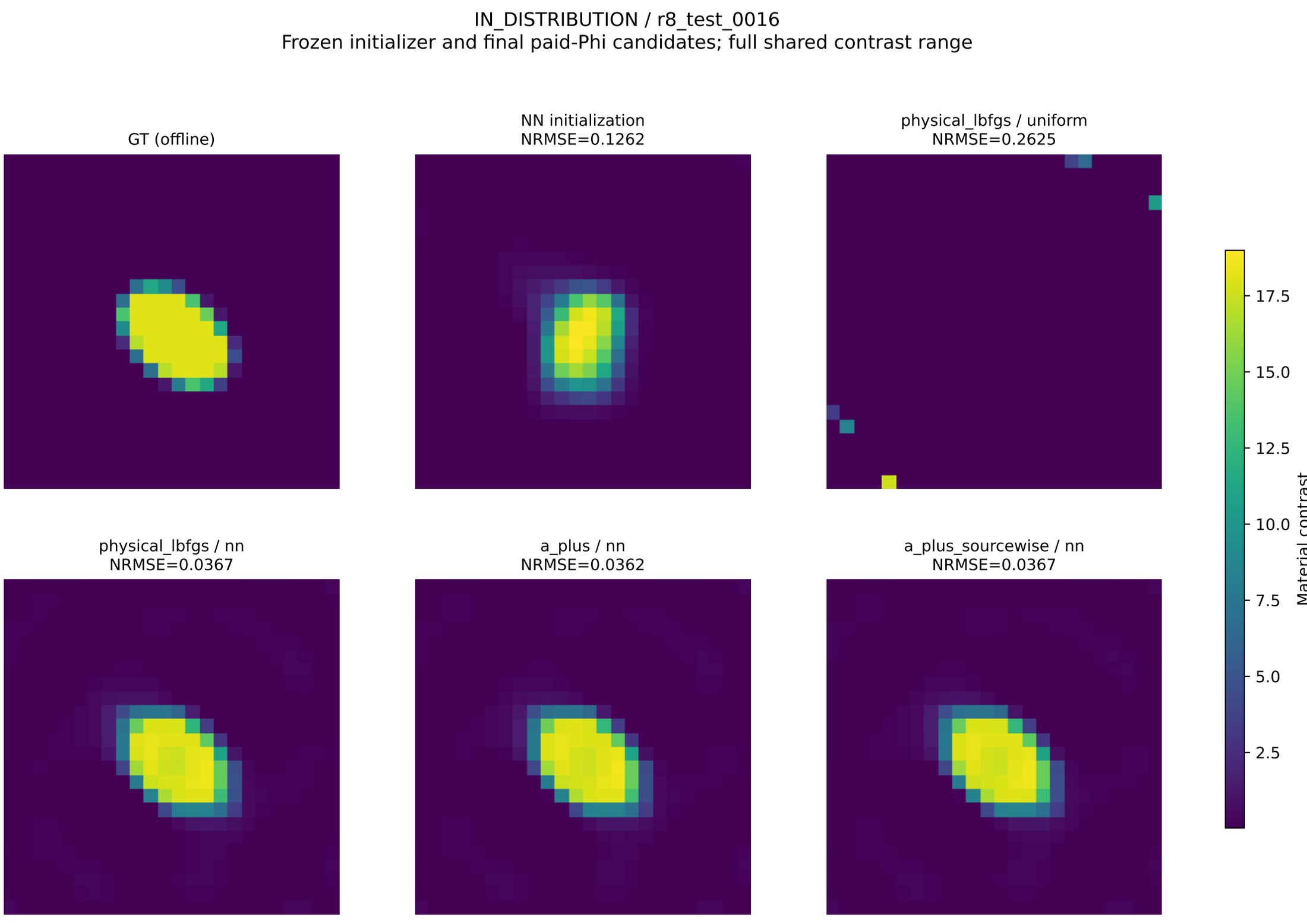

IN_DISTRIBUTION / r8_test_0011
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1926
physical_lbfgs / uniform
NRMSE=0.3228
physical_lbfgs / nn
NRMSE=0.0796
a_plus / nn
NRMSE=0.2880
a_plus_sourcewise / nn
NRMSE=0.2888
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

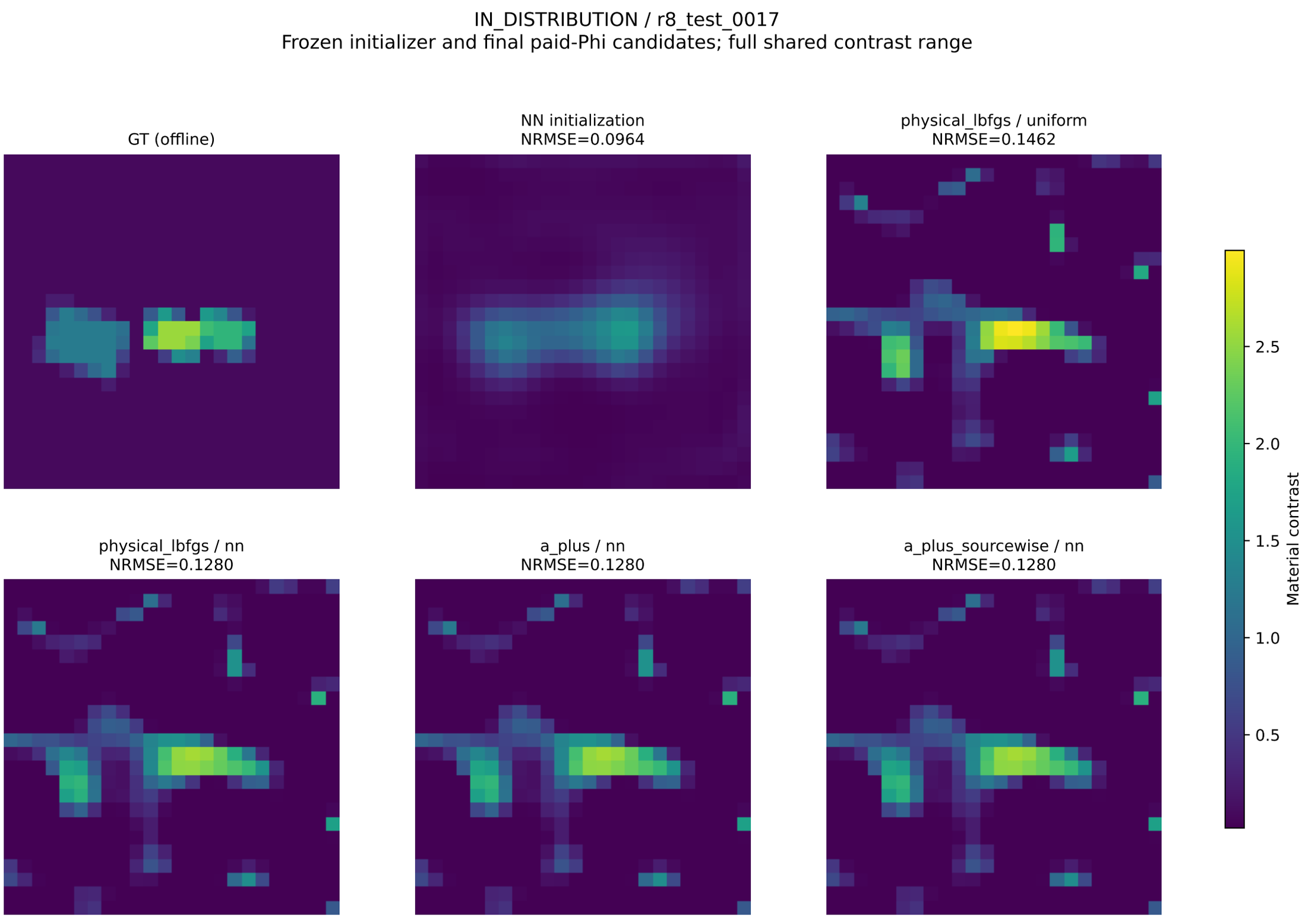

IN_DISTRIBUTION / r8_test_0012
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1229
physical_lbfgs / uniform
NRMSE=0.1401
physical_lbfgs / nn
NRMSE=0.1364
a_plus / nn
NRMSE=1.9363
a_plus_sourcewise / nn
NRMSE=1.9364
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

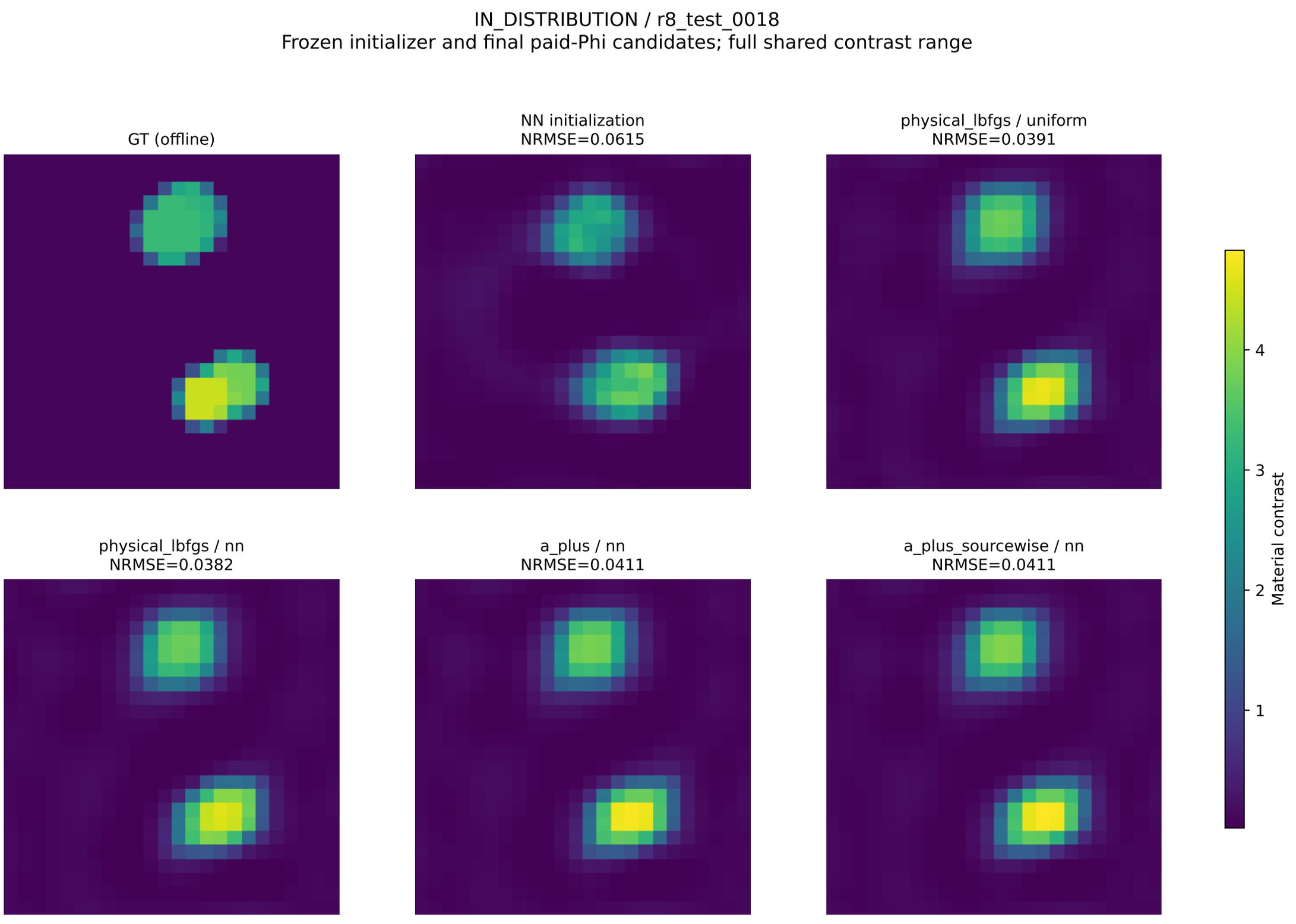

IN_DISTRIBUTION / r8_test_0013
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1798
physical_lbfgs / uniform
NRMSE=0.3302
physical_lbfgs / nn
NRMSE=0.0644
a_plus / nn
NRMSE=0.3749
a_plus_sourcewise / nn
NRMSE=0.3651
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

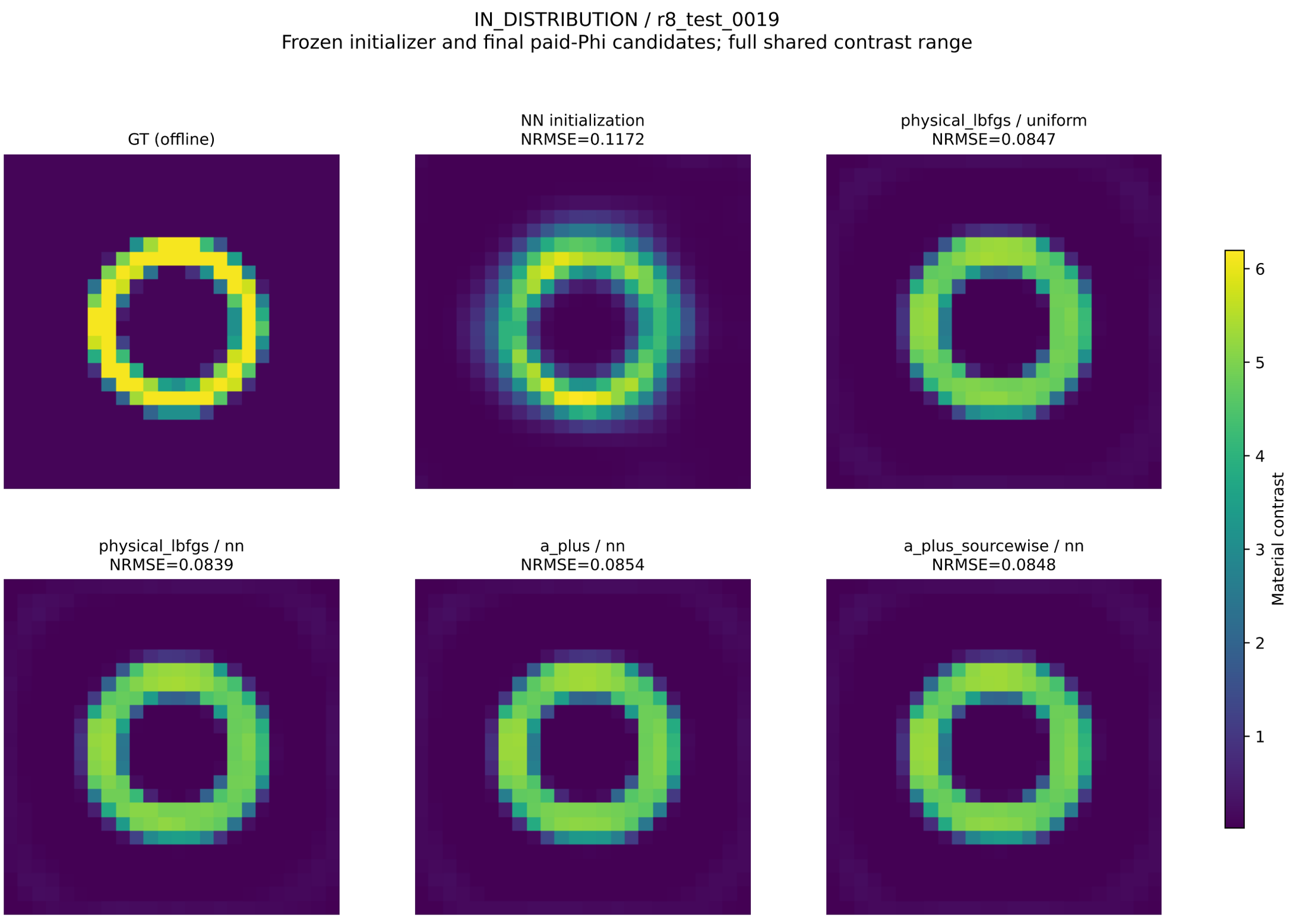

IN_DISTRIBUTION / r8_test_0014
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0934
physical_lbfgs / uniform
NRMSE=0.0627
physical_lbfgs / nn
NRMSE=0.0600
a_plus / nn
NRMSE=0.0633
a_plus_sourcewise / nn
NRMSE=0.0642
Material contrast
1
2
3
4
5
6

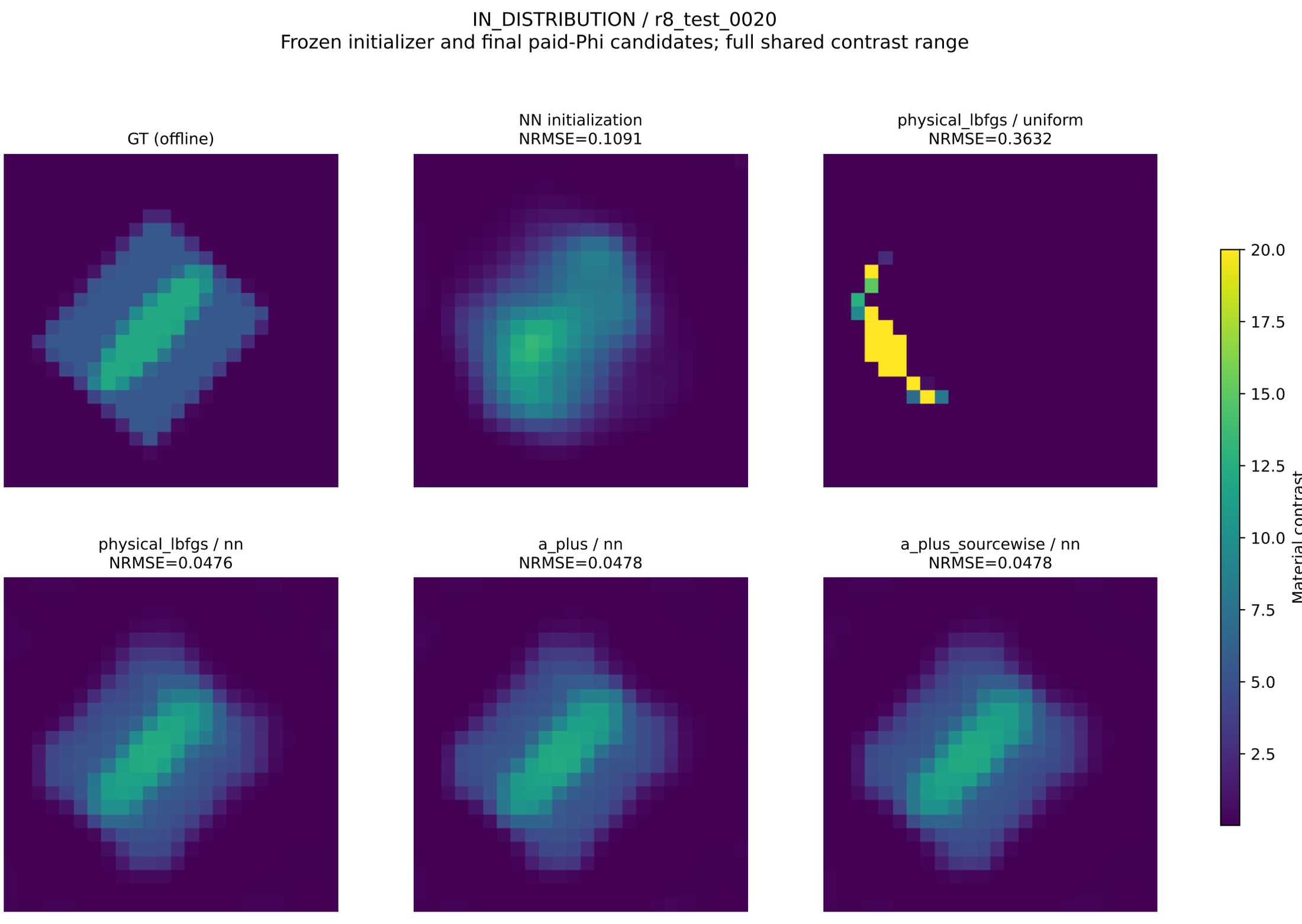

IN_DISTRIBUTION / r8_test_0015
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0787
physical_lbfgs / uniform
NRMSE=0.0232
physical_lbfgs / nn
NRMSE=0.0246
a_plus / nn
NRMSE=0.0240
a_plus_sourcewise / nn
NRMSE=0.0236
Material contrast
12
10
8
6
4
2

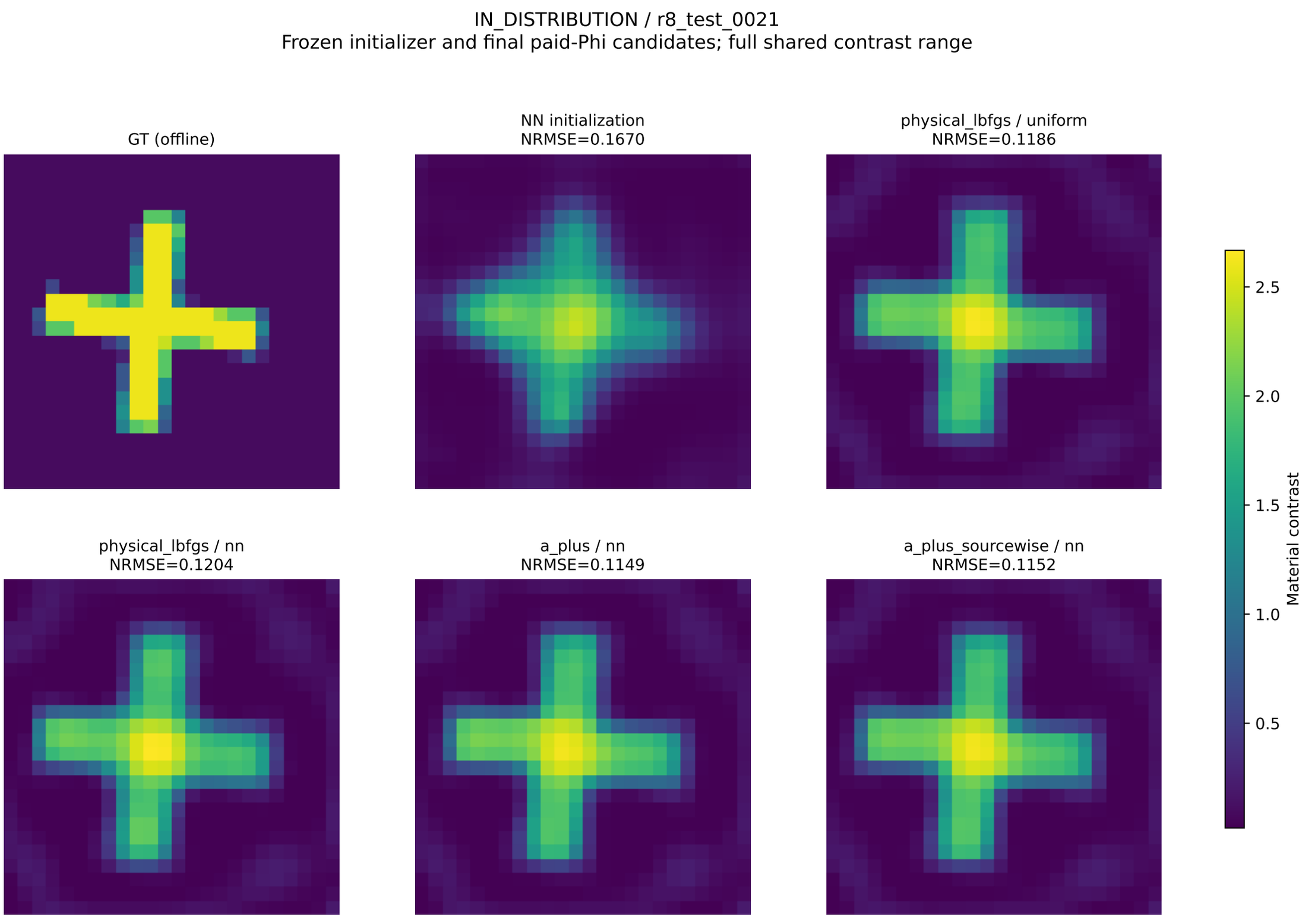

IN_DISTRIBUTION / r8_test_0016
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1262
physical_lbfgs / uniform
NRMSE=0.2625
physical_lbfgs / nn
NRMSE=0.0367
a_plus / nn
NRMSE=0.0362
a_plus_sourcewise / nn
NRMSE=0.0367
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

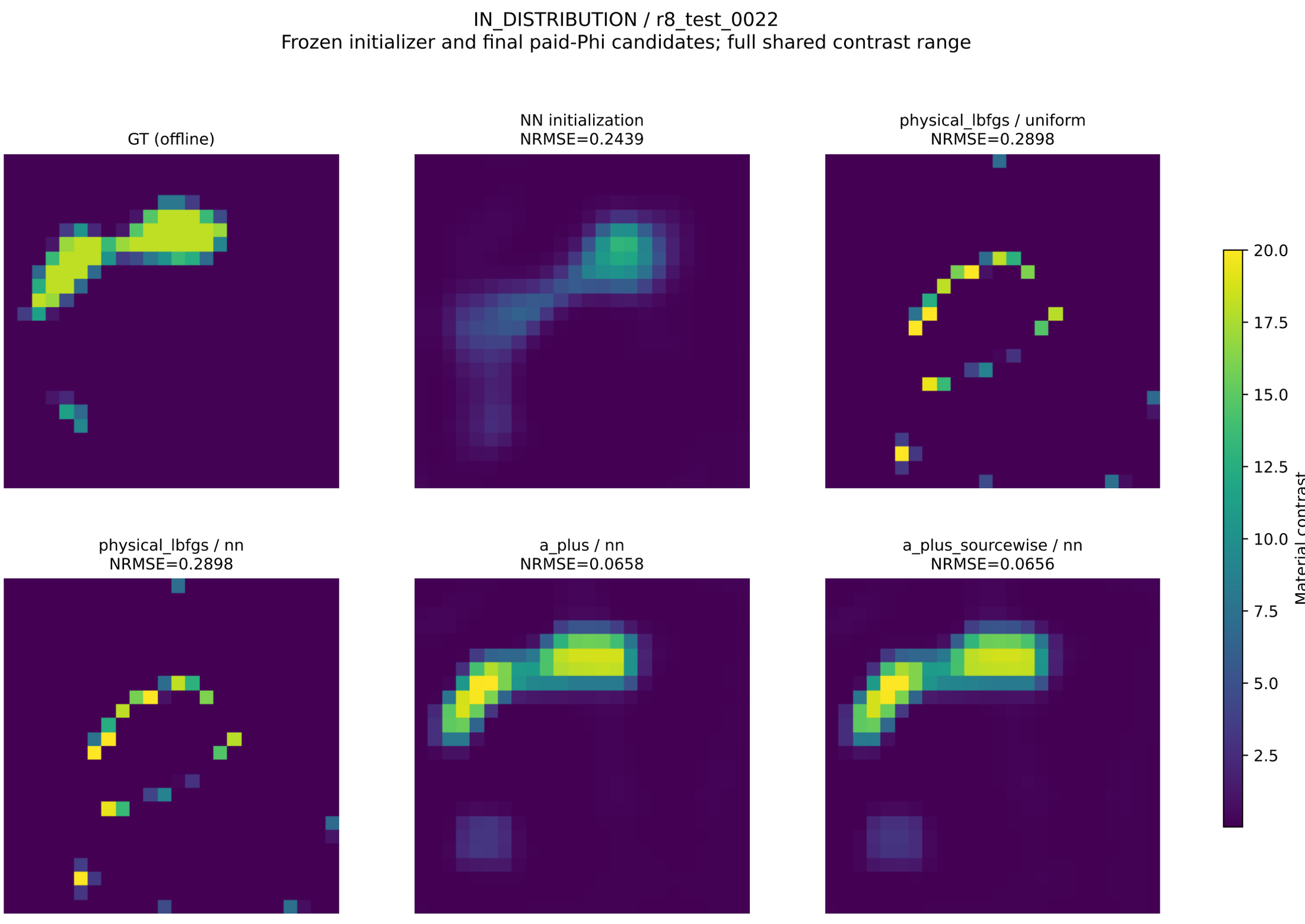

IN_DISTRIBUTION / r8_test_0017
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0964
physical_lbfgs / uniform
NRMSE=0.1462
physical_lbfgs / nn
NRMSE=0.1280
a_plus / nn
NRMSE=0.1280
a_plus_sourcewise / nn
NRMSE=0.1280
Material contrast
2.5
2.0
1.5
1.0
0.5

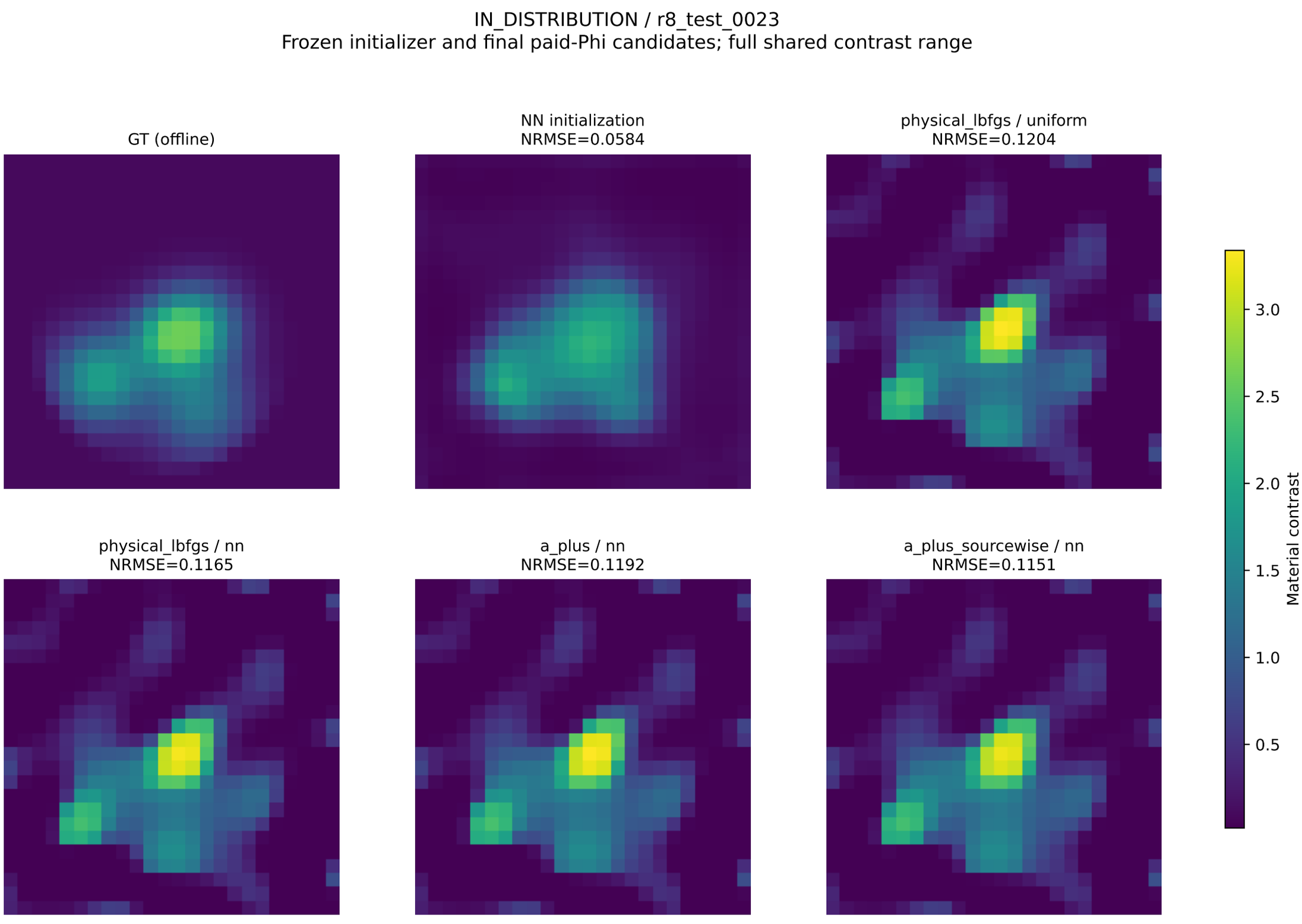

IN_DISTRIBUTION / r8_test_0018
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0615
physical_lbfgs / uniform
NRMSE=0.0391
physical_lbfgs / nn
NRMSE=0.0382
a_plus / nn
NRMSE=0.0411
a_plus_sourcewise / nn
NRMSE=0.0411
Material contrast
1
2
3
4

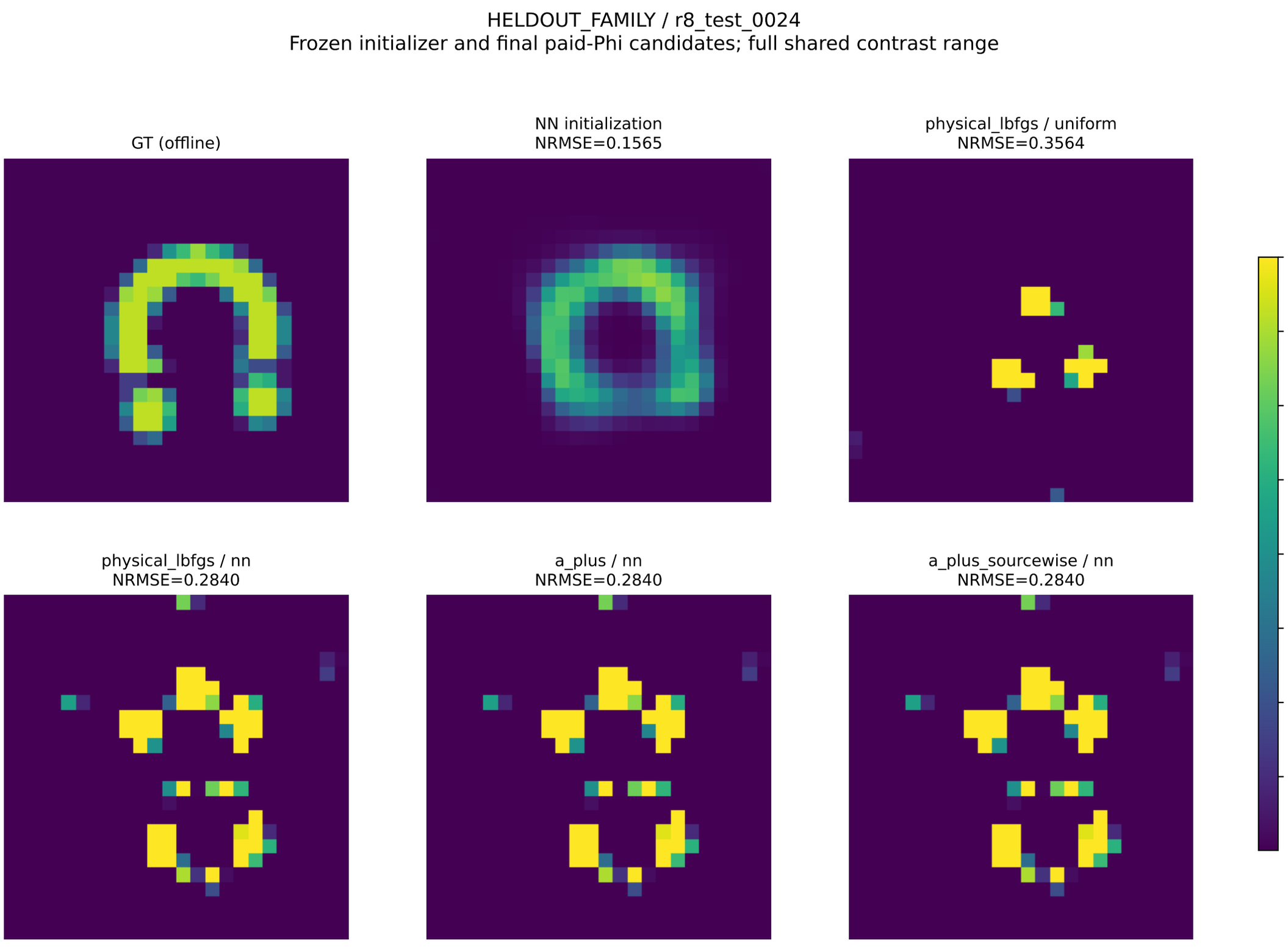

IN_DISTRIBUTION / r8_test_0019
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1172
physical_lbfgs / uniform
NRMSE=0.0847
physical_lbfgs / nn
NRMSE=0.0839
a_plus / nn
NRMSE=0.0854
a_plus_sourcewise / nn
NRMSE=0.0848
Material contrast
1
2
3
4
5
6

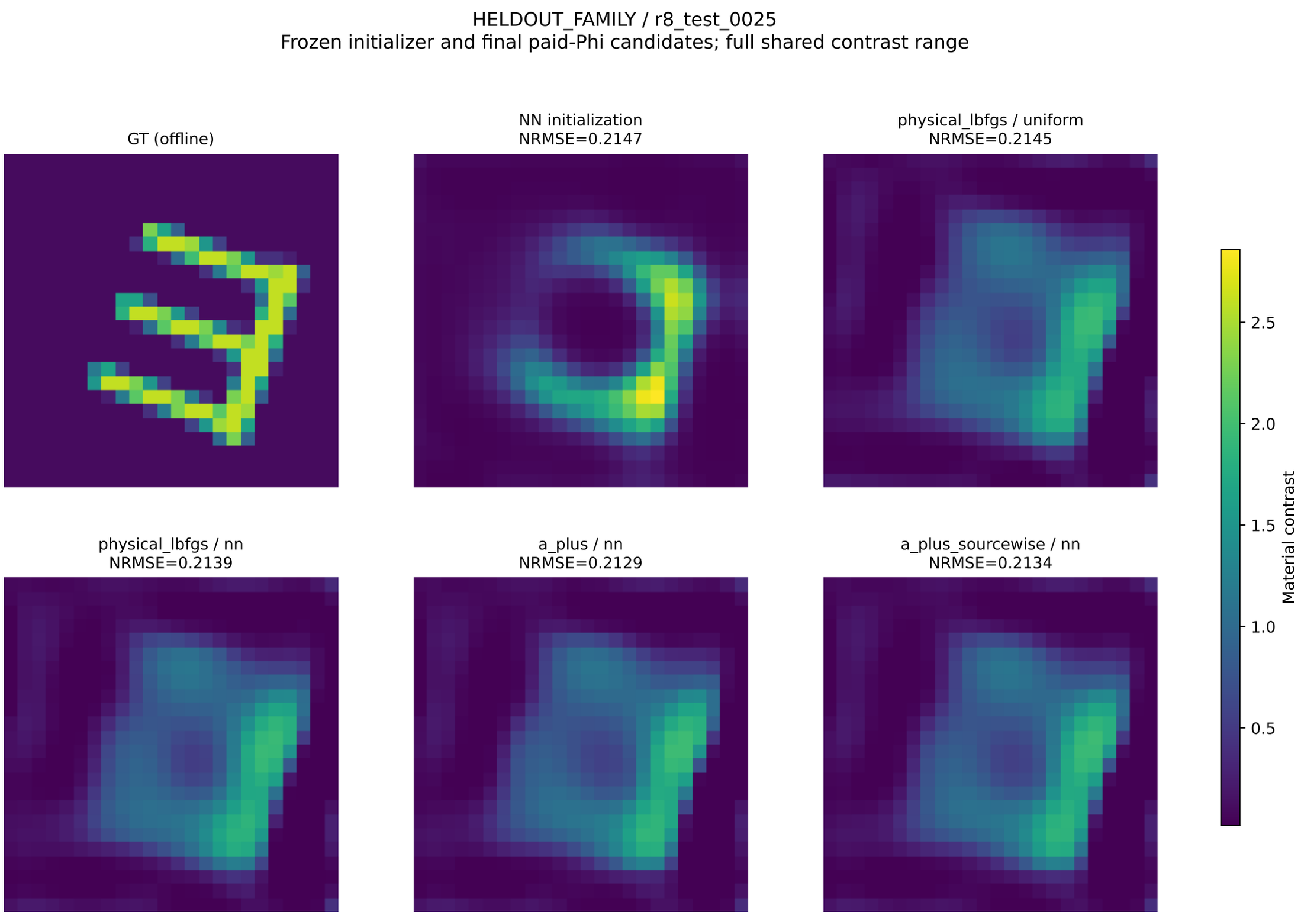

IN_DISTRIBUTION / r8_test_0020
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1091
physical_lbfgs / uniform
NRMSE=0.3632
physical_lbfgs / nn
NRMSE=0.0476
a_plus / nn
NRMSE=0.0478
a_plus_sourcewise / nn
NRMSE=0.0478
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

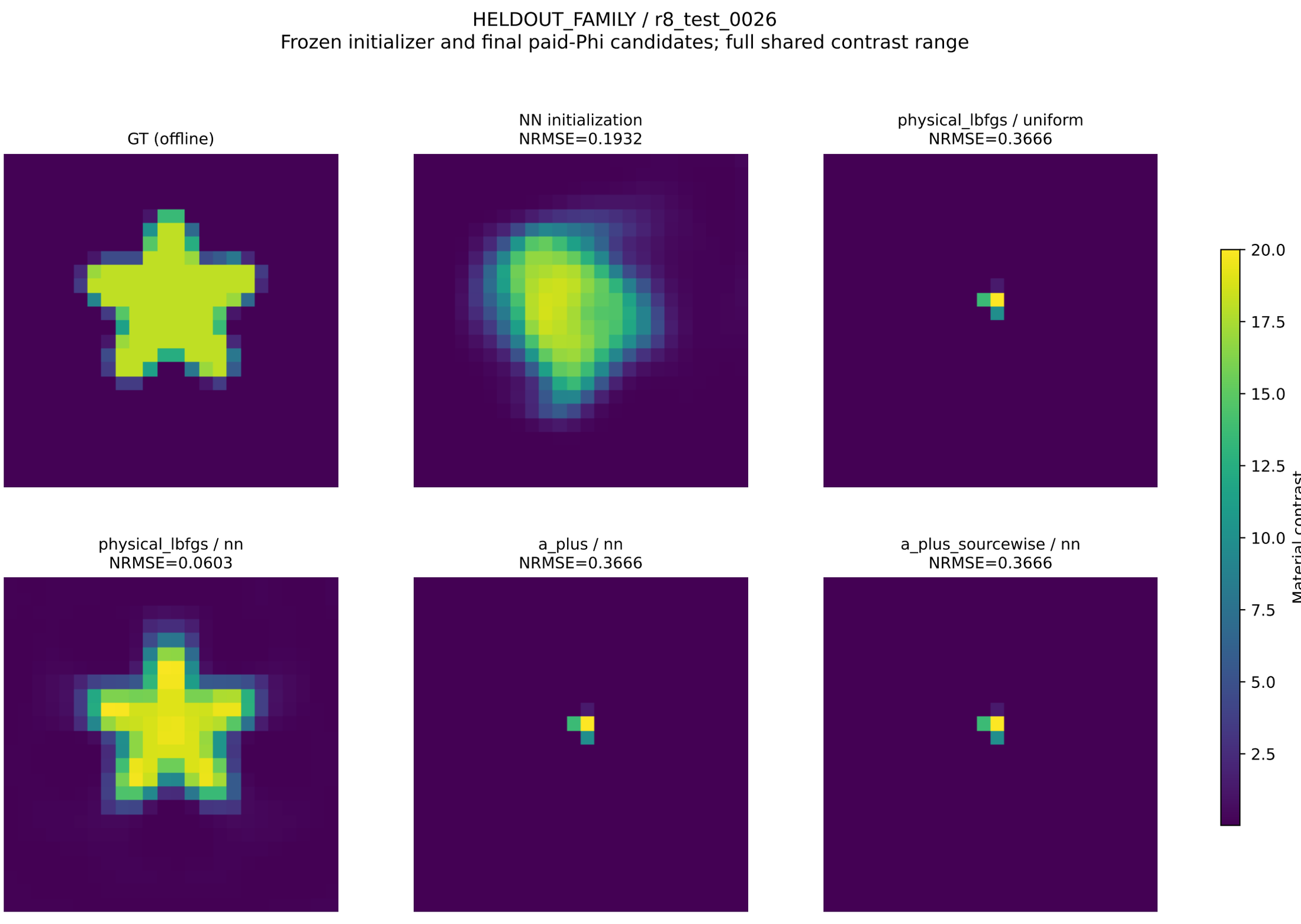

IN_DISTRIBUTION / r8_test_0021
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1670
physical_lbfgs / uniform
NRMSE=0.1186
physical_lbfgs / nn
NRMSE=0.1204
a_plus / nn
NRMSE=0.1149
a_plus_sourcewise / nn
NRMSE=0.1152
Material contrast
2.5
2.0
1.5
1.0
0.5

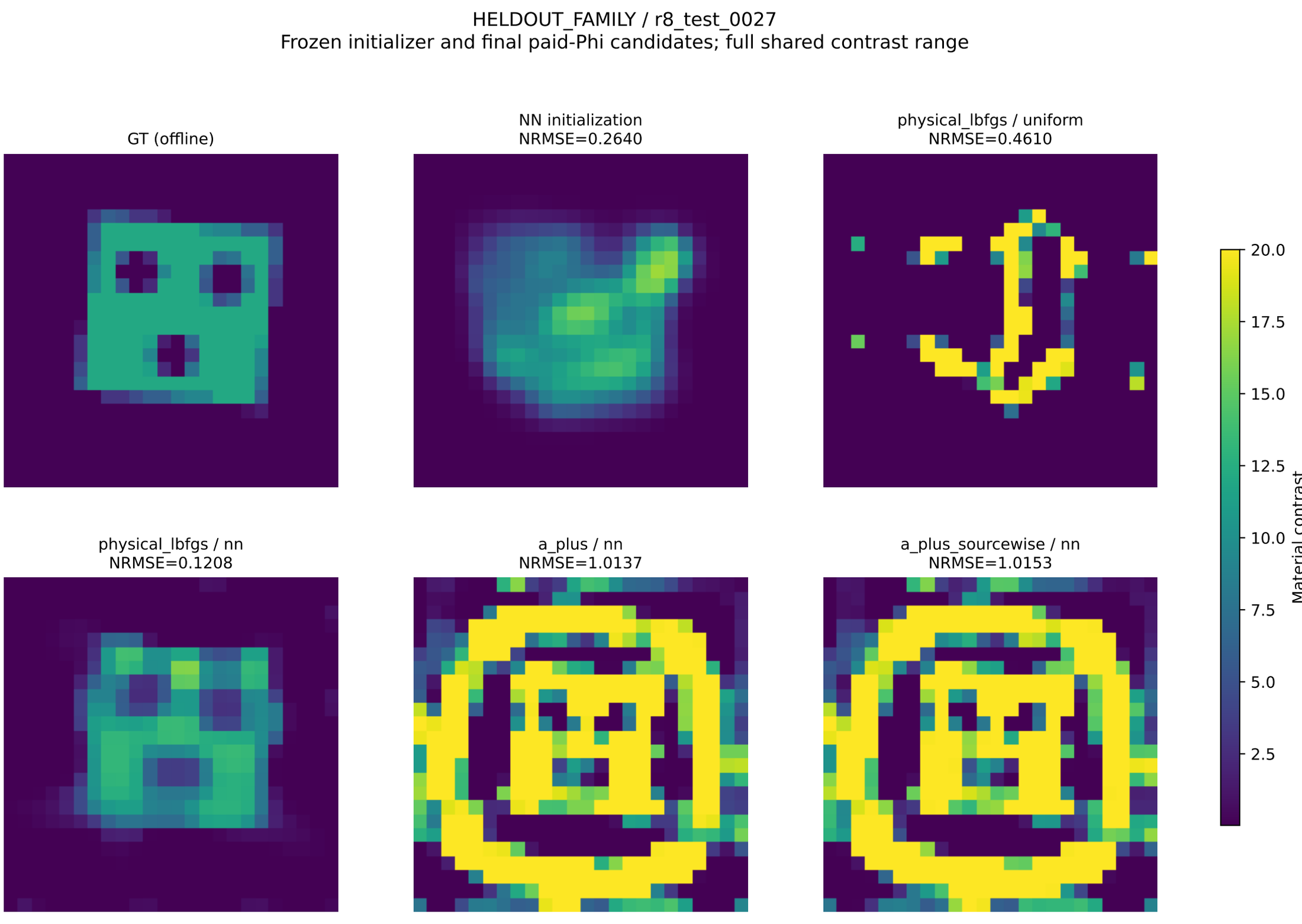

IN_DISTRIBUTION / r8_test_0022
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2439
physical_lbfgs / uniform
NRMSE=0.2898
physical_lbfgs / nn
NRMSE=0.2898
a_plus / nn
NRMSE=0.0658
a_plus_sourcewise / nn
NRMSE=0.0656
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

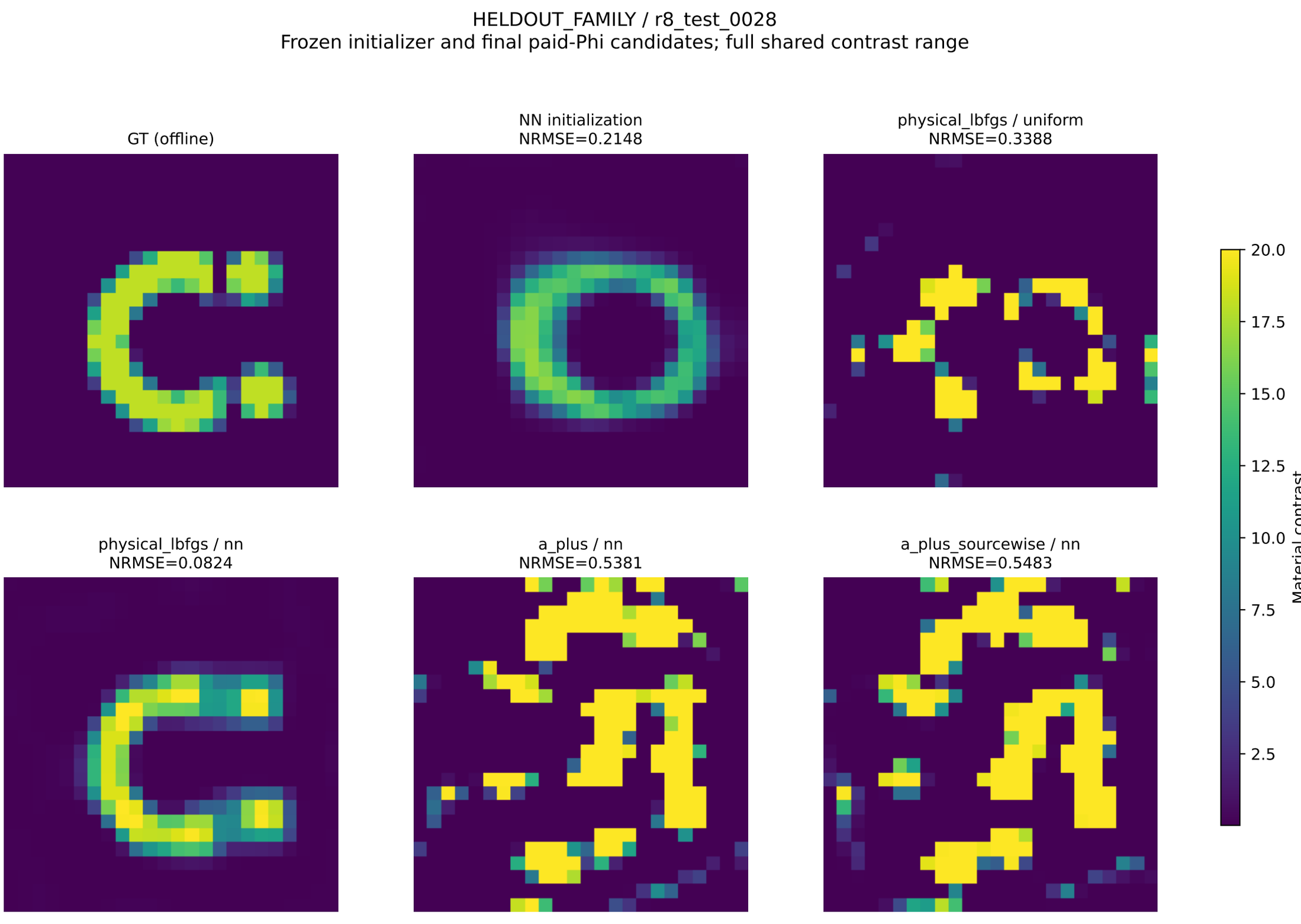

IN_DISTRIBUTION / r8_test_0023
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0584
physical_lbfgs / uniform
NRMSE=0.1204
physical_lbfgs / nn
NRMSE=0.1165
a_plus / nn
NRMSE=0.1192
a_plus_sourcewise / nn
NRMSE=0.1151
Material contrast
3.0
2.5
2.0
1.5
1.0
0.5

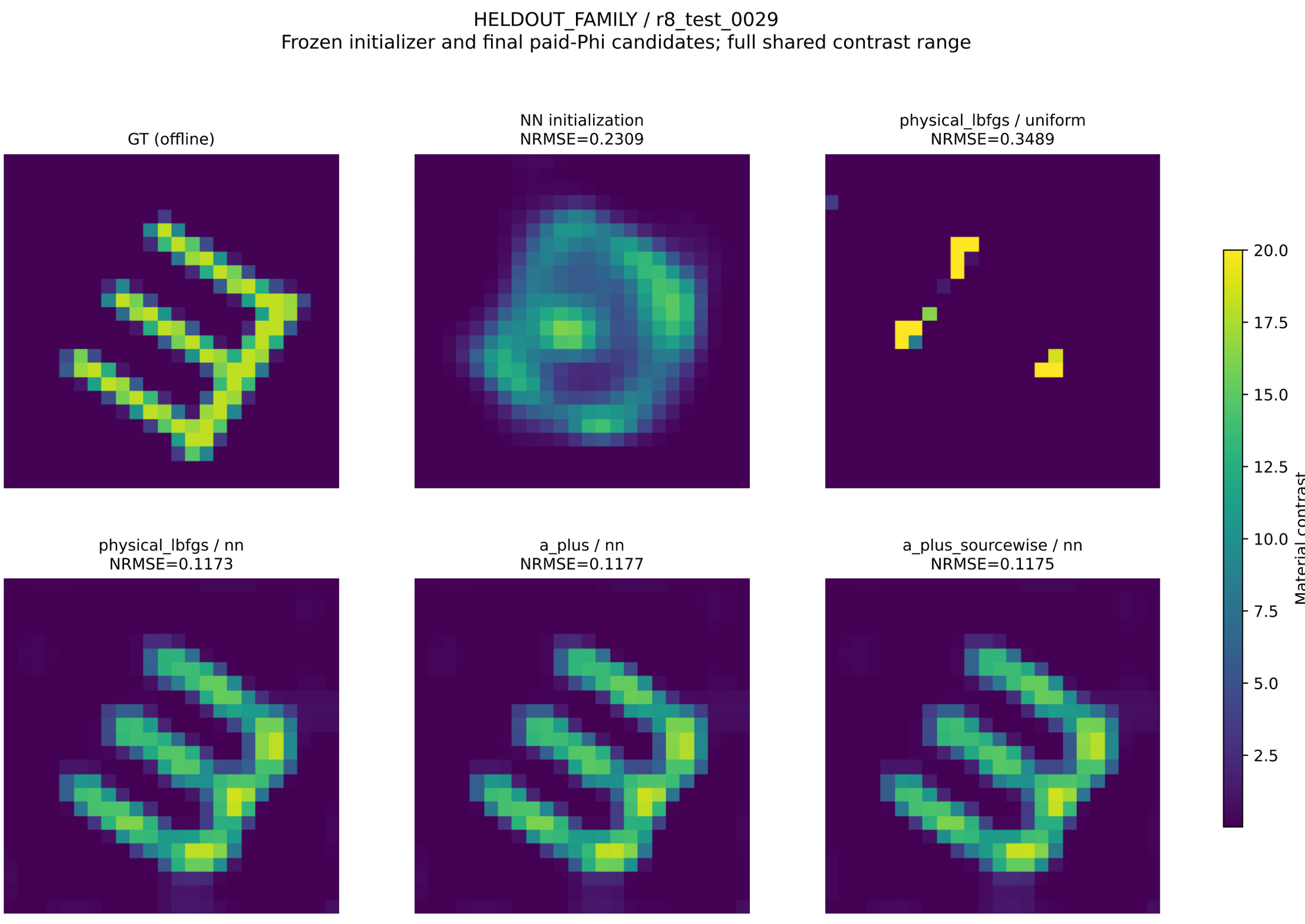

HELDOUT_FAMILY / r8_test_0024
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1565
physical_lbfgs / uniform
NRMSE=0.3564
physical_lbfgs / nn
NRMSE=0.2840
a_plus / nn
NRMSE=0.2840
a_plus_sourcewise / nn
NRMSE=0.2840
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

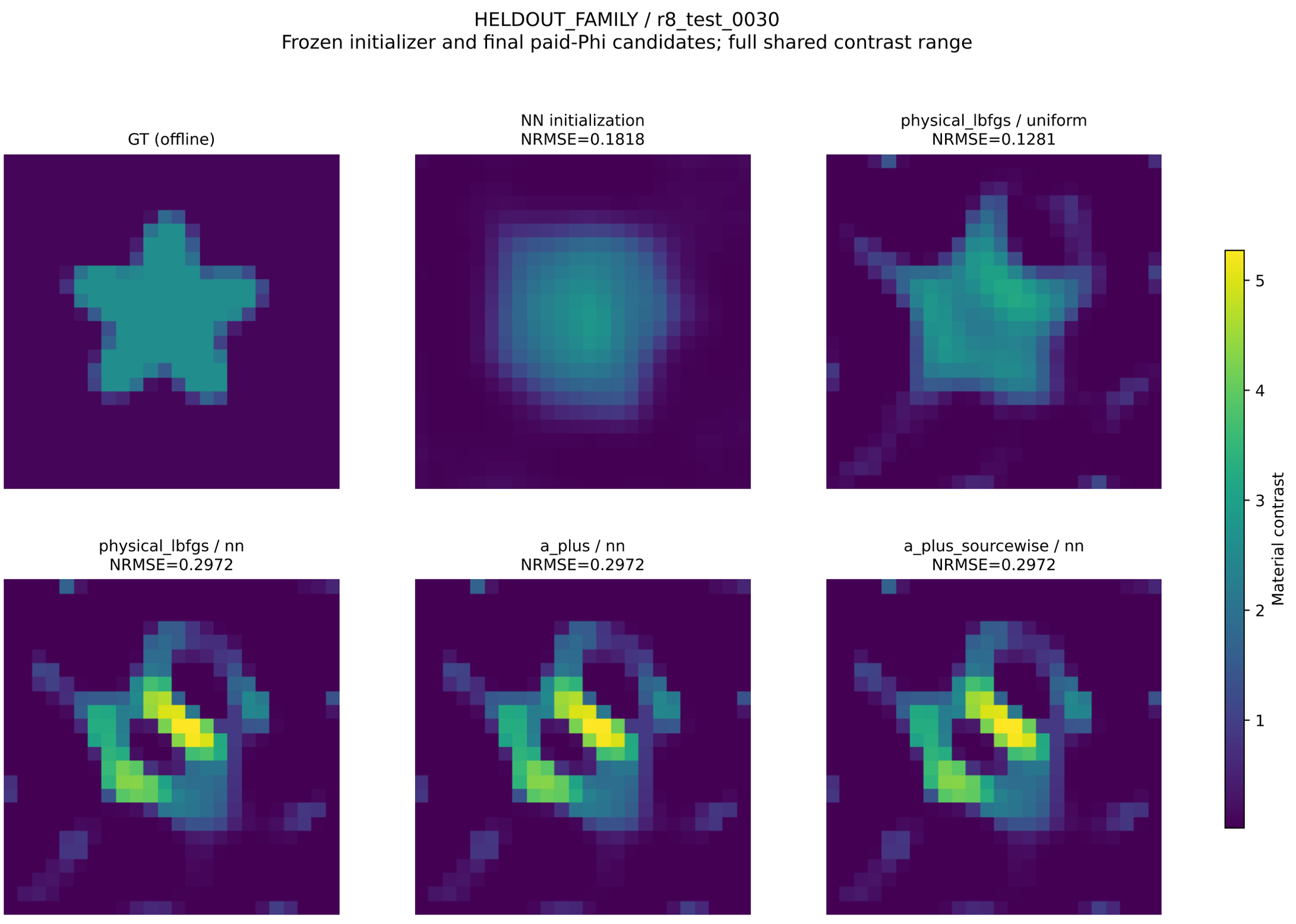

HELDOUT_FAMILY / r8_test_0025
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2147
physical_lbfgs / uniform
NRMSE=0.2145
physical_lbfgs / nn
NRMSE=0.2139
a_plus / nn
NRMSE=0.2129
a_plus_sourcewise / nn
NRMSE=0.2134
Material contrast
0.5
1.0
1.5
2.0
2.5

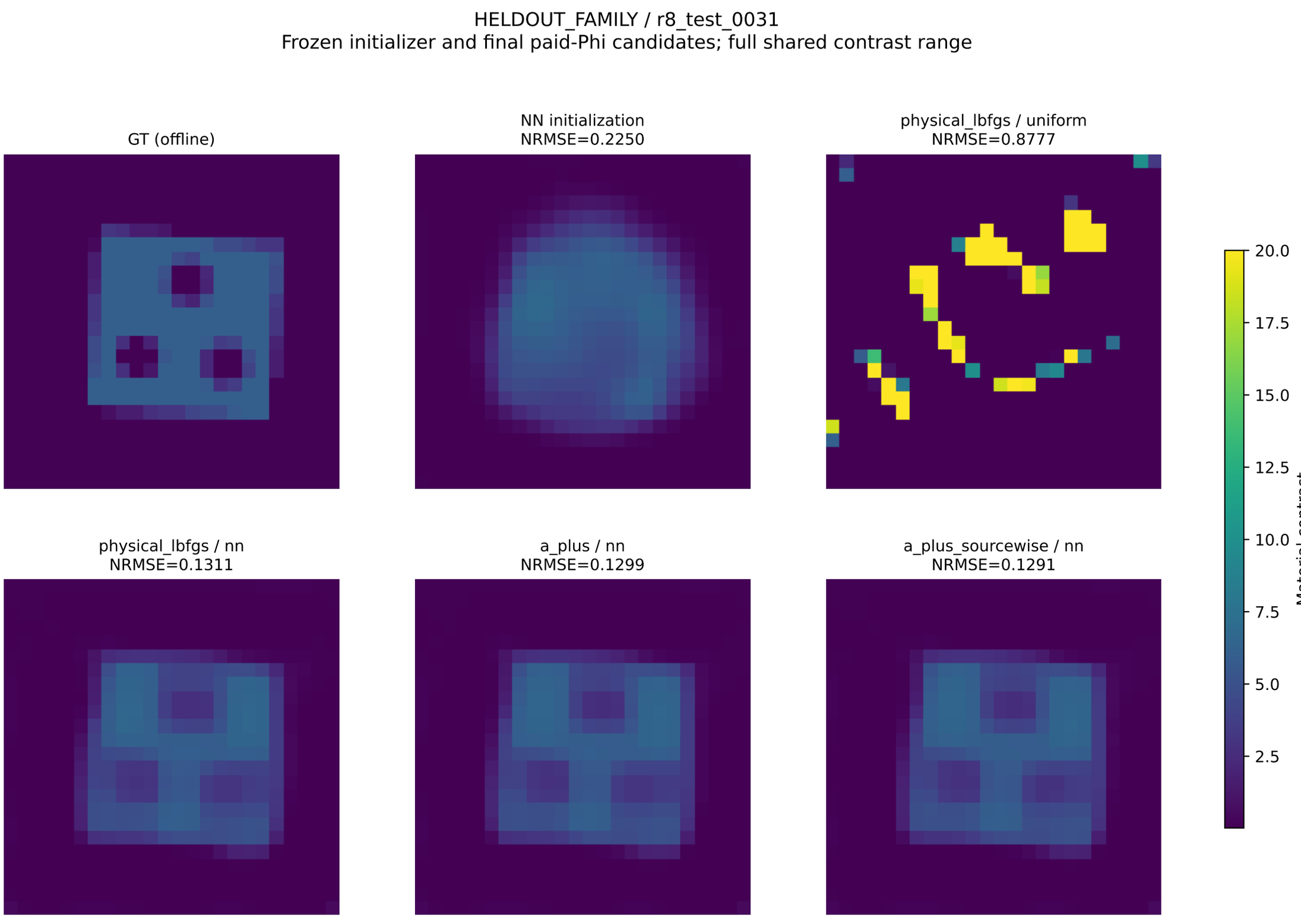

HELDOUT_FAMILY / r8_test_0026
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1932
physical_lbfgs / uniform
NRMSE=0.3666
physical_lbfgs / nn
NRMSE=0.0603
a_plus / nn
NRMSE=0.3666
a_plus_sourcewise / nn
NRMSE=0.3666
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

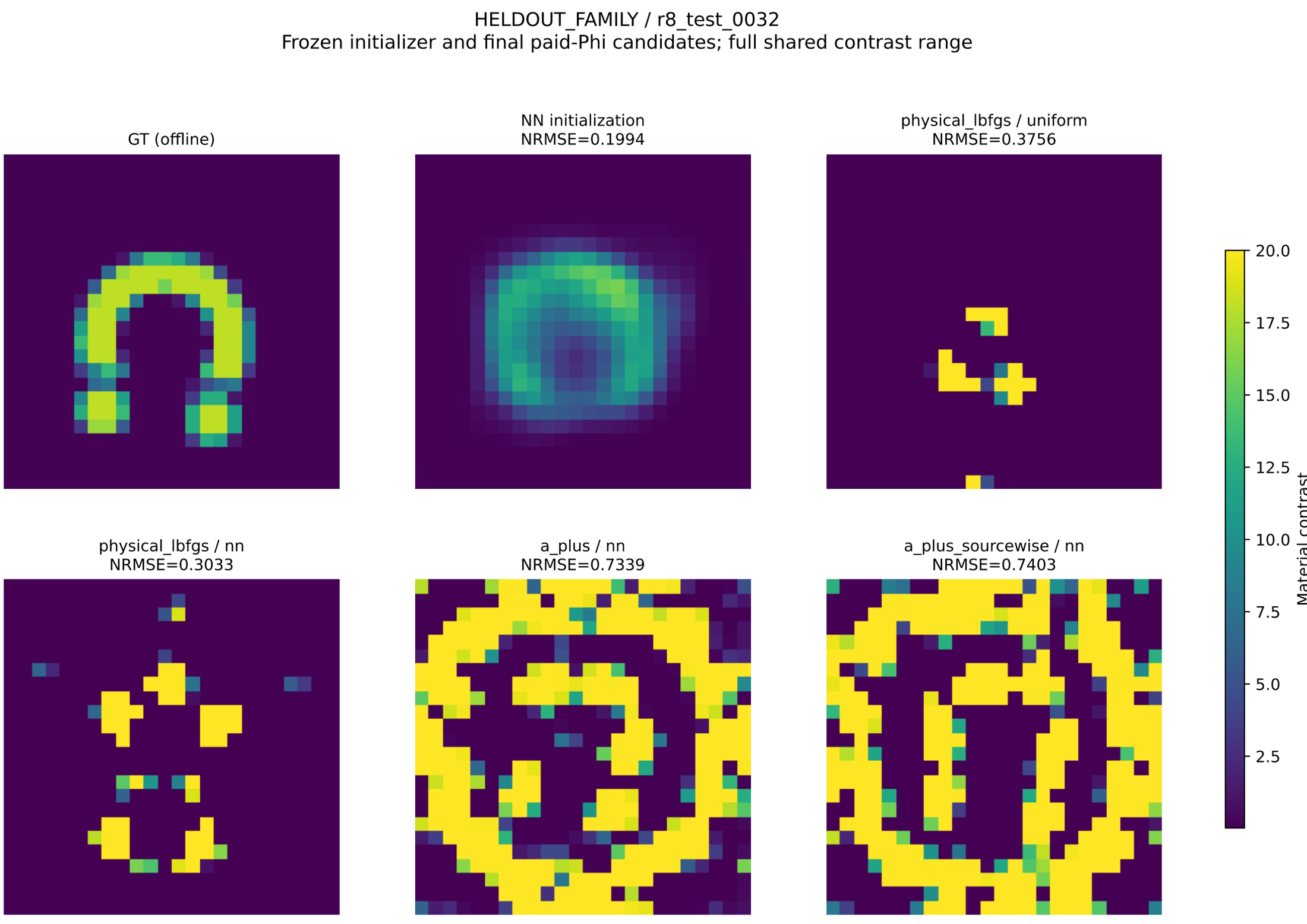

HELDOUT_FAMILY / r8_test_0027
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2640
physical_lbfgs / uniform
NRMSE=0.4610
physical_lbfgs / nn
NRMSE=0.1208
a_plus / nn
NRMSE=1.0137
a_plus_sourcewise / nn
NRMSE=1.0153
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

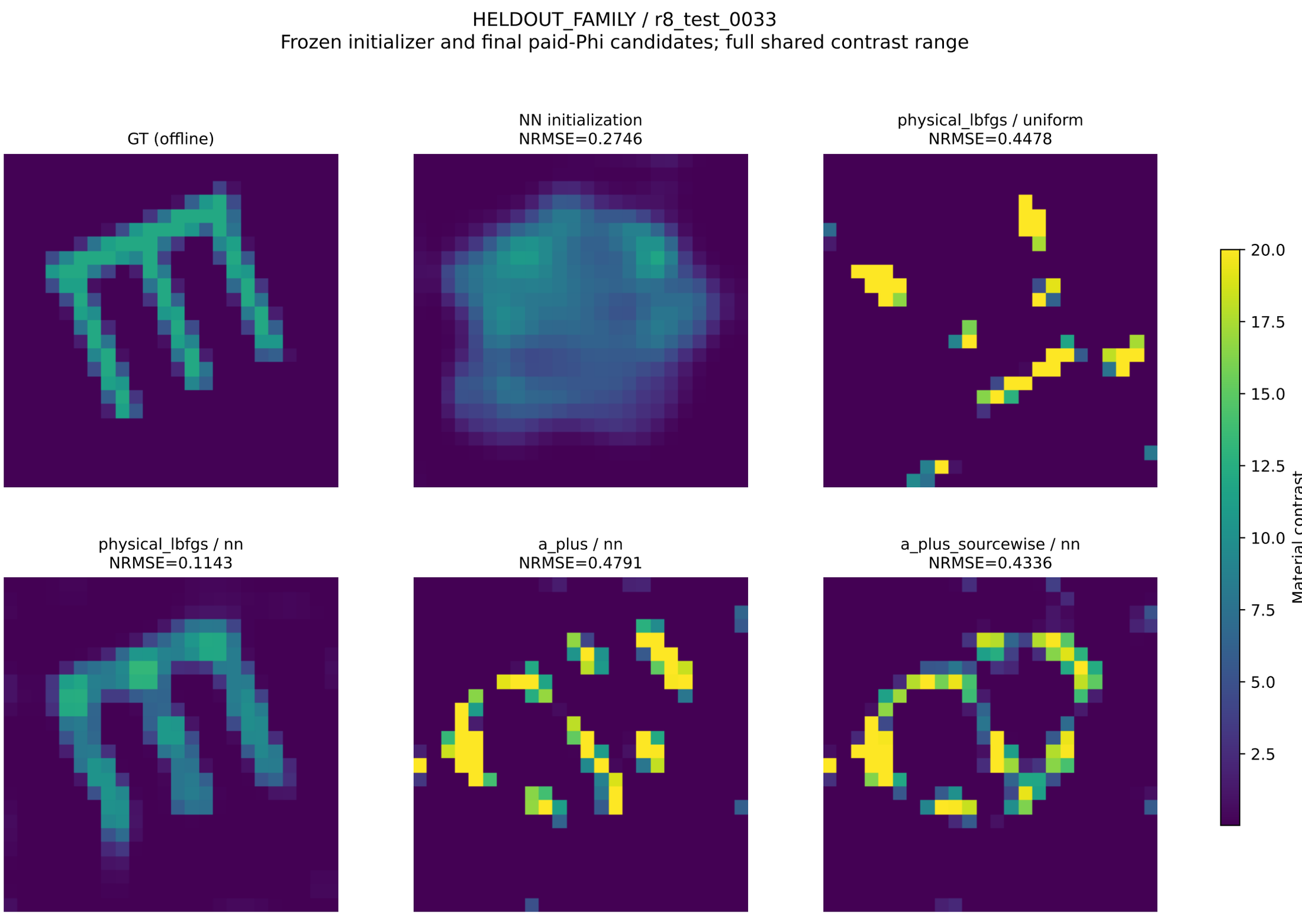

HELDOUT_FAMILY / r8_test_0028
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2148
physical_lbfgs / uniform
NRMSE=0.3388
physical_lbfgs / nn
NRMSE=0.0824
a_plus / nn
NRMSE=0.5381
a_plus_sourcewise / nn
NRMSE=0.5483
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

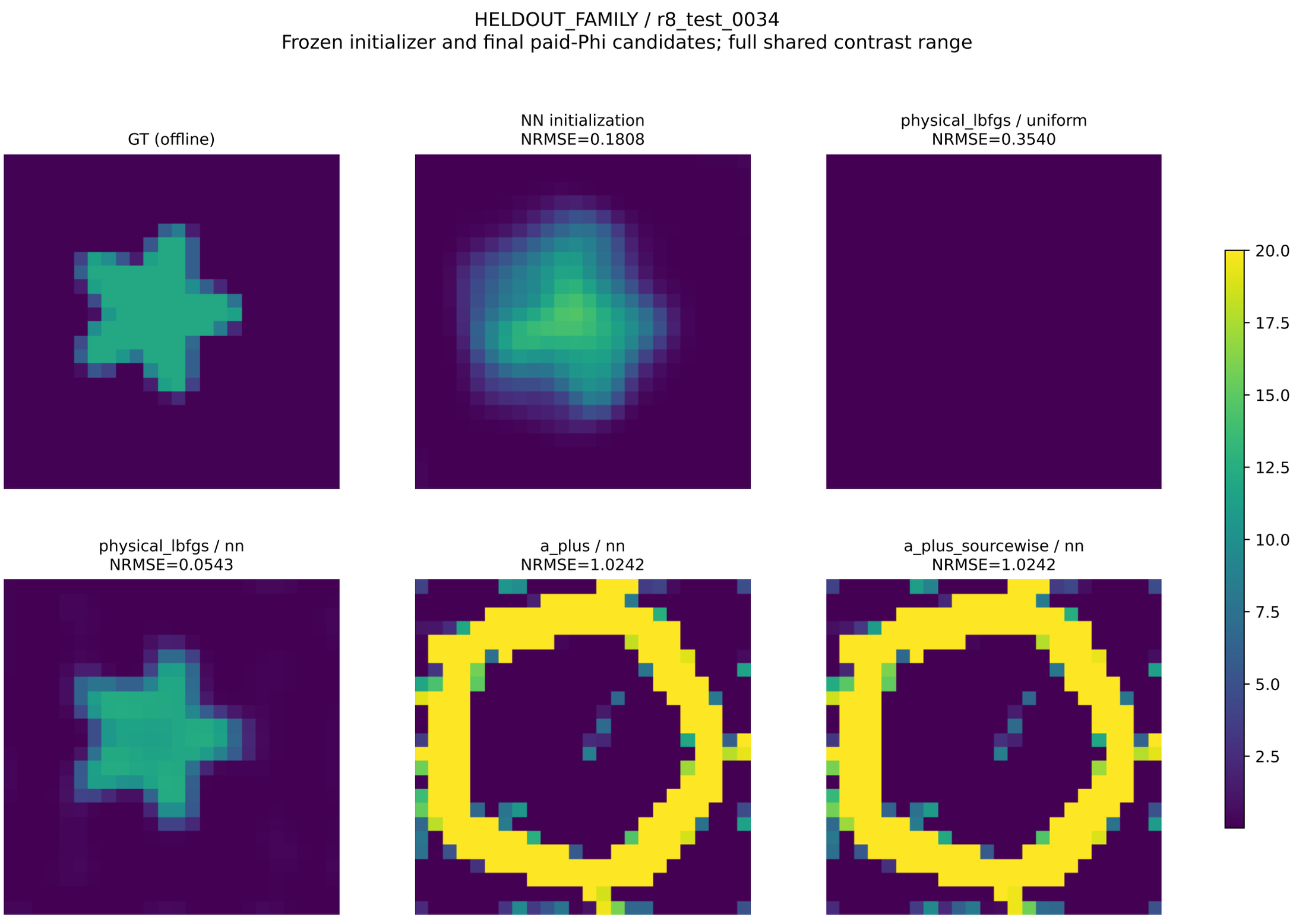

HELDOUT_FAMILY / r8_test_0029
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2309
physical_lbfgs / uniform
NRMSE=0.3489
physical_lbfgs / nn
NRMSE=0.1173
a_plus / nn
NRMSE=0.1177
a_plus_sourcewise / nn
NRMSE=0.1175
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

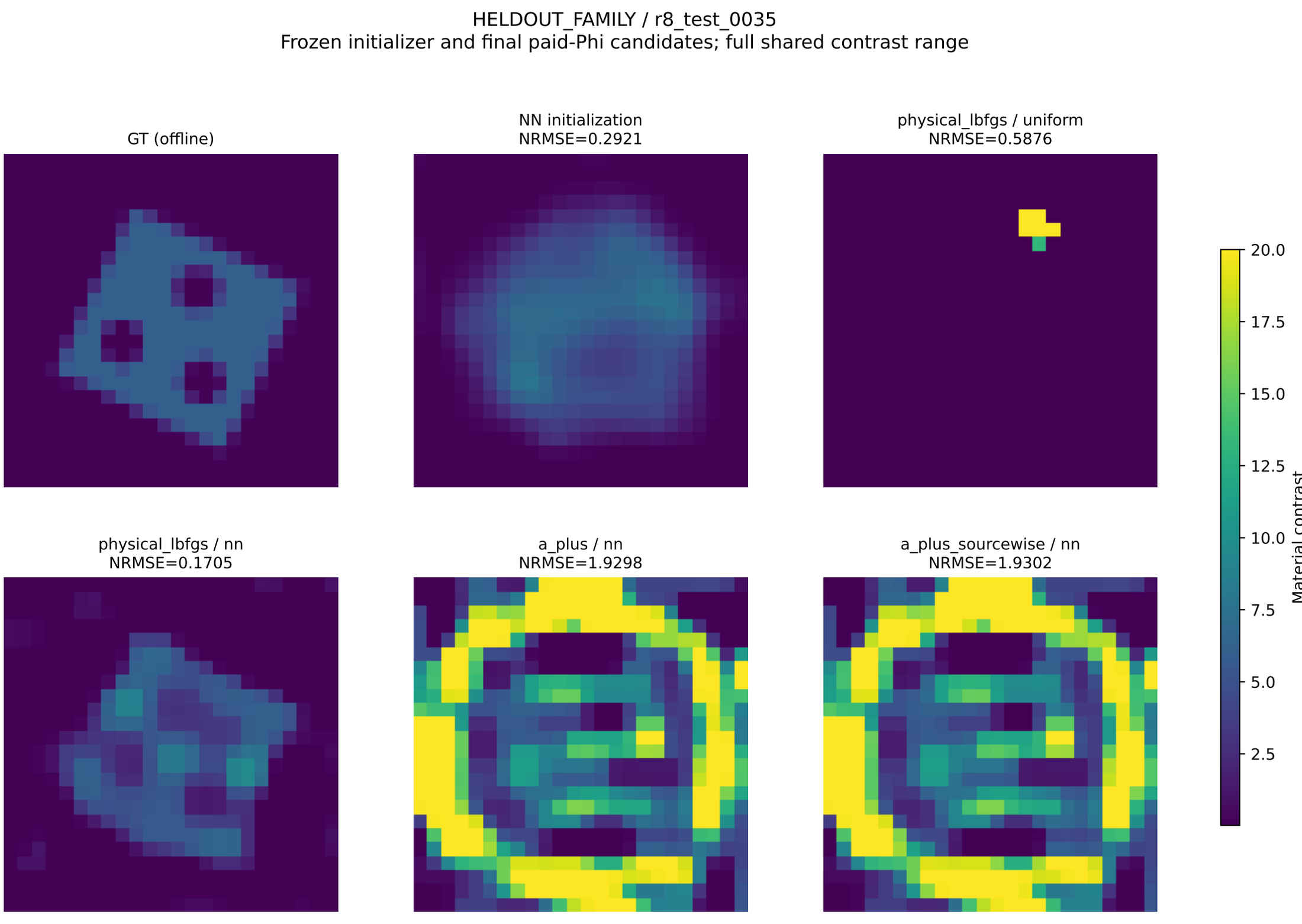

HELDOUT_FAMILY / r8_test_0030
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1818
physical_lbfgs / uniform
NRMSE=0.1281
physical_lbfgs / nn
NRMSE=0.2972
a_plus / nn
NRMSE=0.2972
a_plus_sourcewise / nn
NRMSE=0.2972
Material contrast
1
2
3
4
5

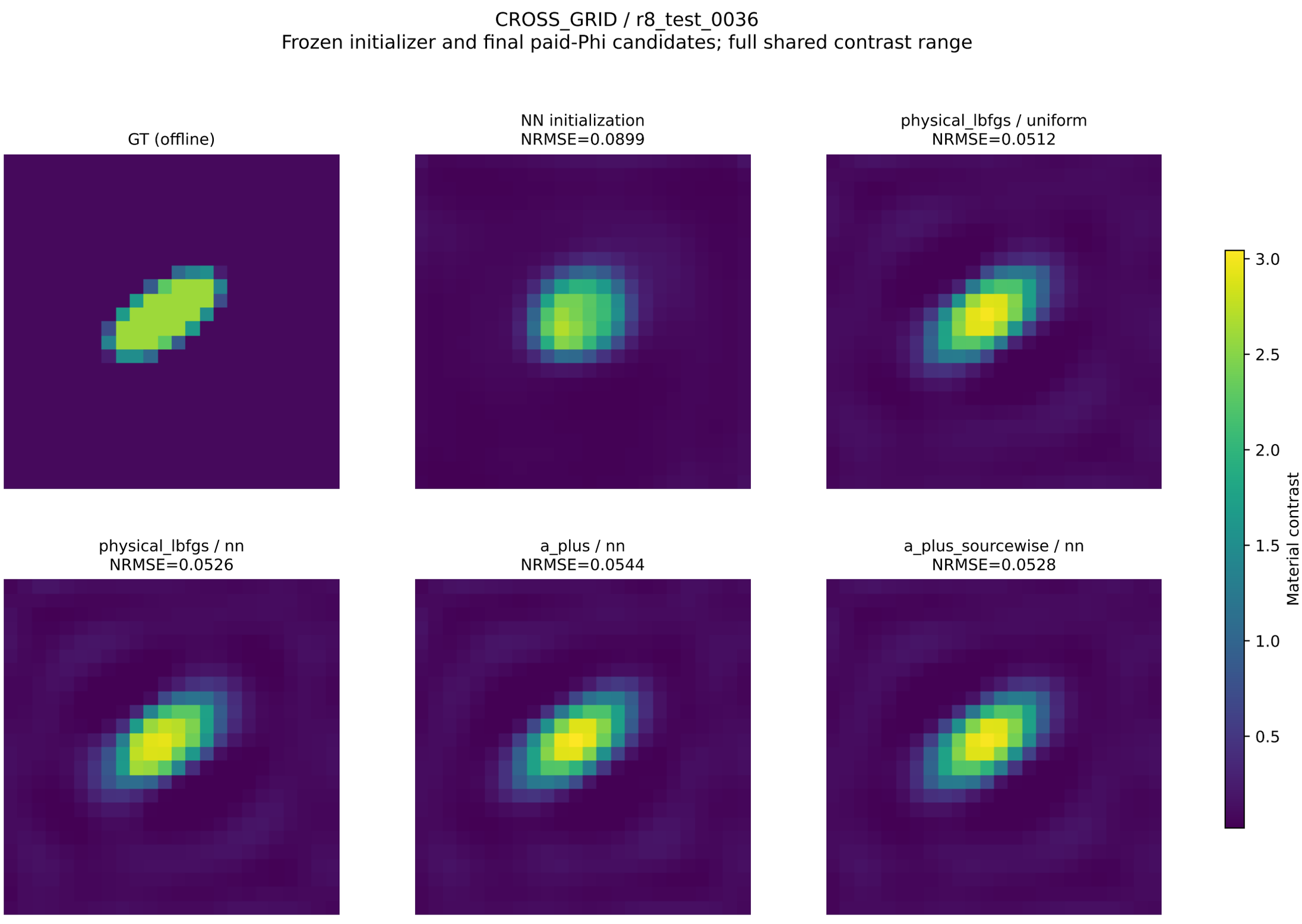

HELDOUT_FAMILY / r8_test_0031
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2250
physical_lbfgs / uniform
NRMSE=0.8777
physical_lbfgs / nn
NRMSE=0.1311
a_plus / nn
NRMSE=0.1299
a_plus_sourcewise / nn
NRMSE=0.1291
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

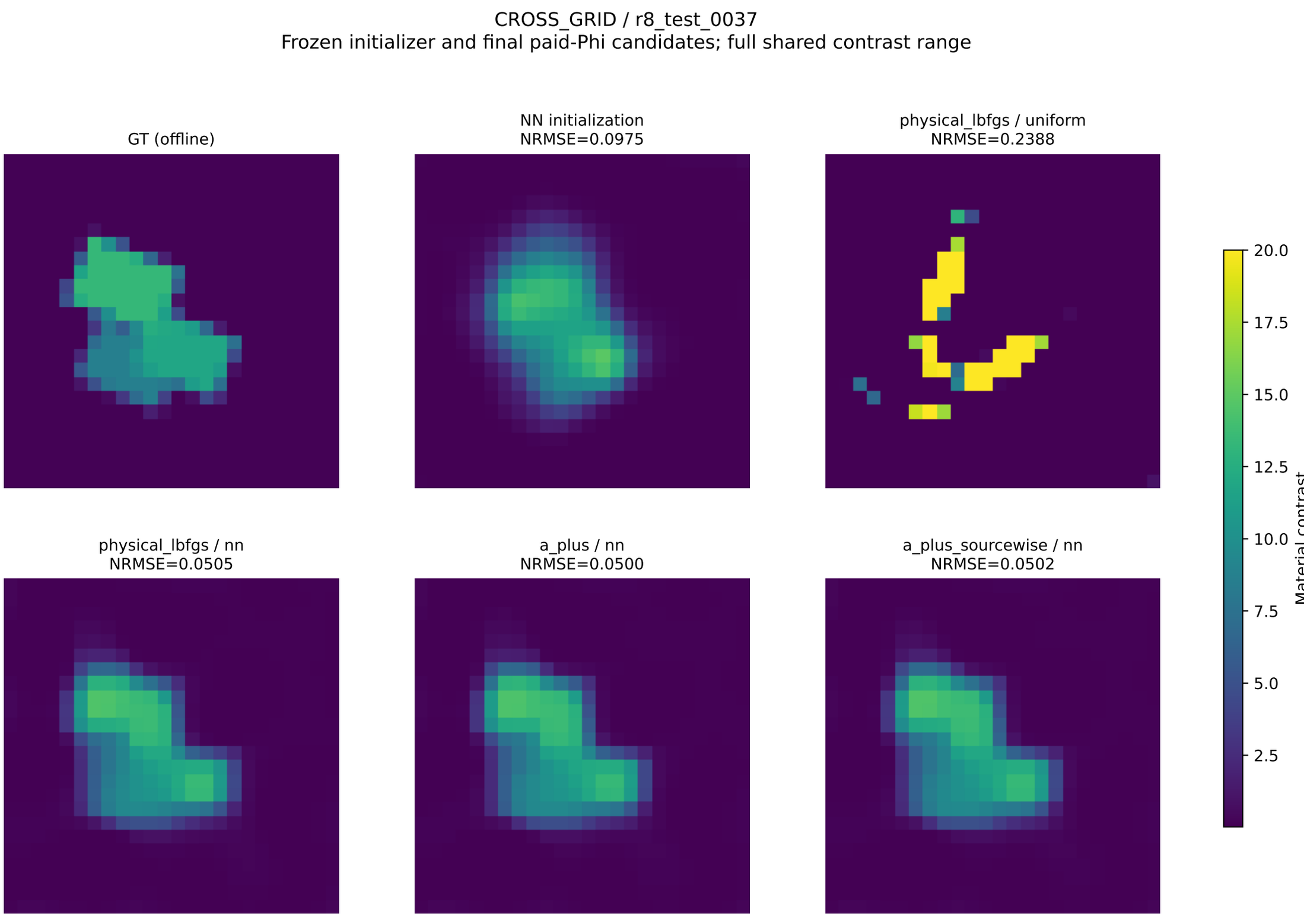

HELDOUT_FAMILY / r8_test_0032
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1994
physical_lbfgs / uniform
NRMSE=0.3756
physical_lbfgs / nn
NRMSE=0.3033
a_plus / nn
NRMSE=0.7339
a_plus_sourcewise / nn
NRMSE=0.7403
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

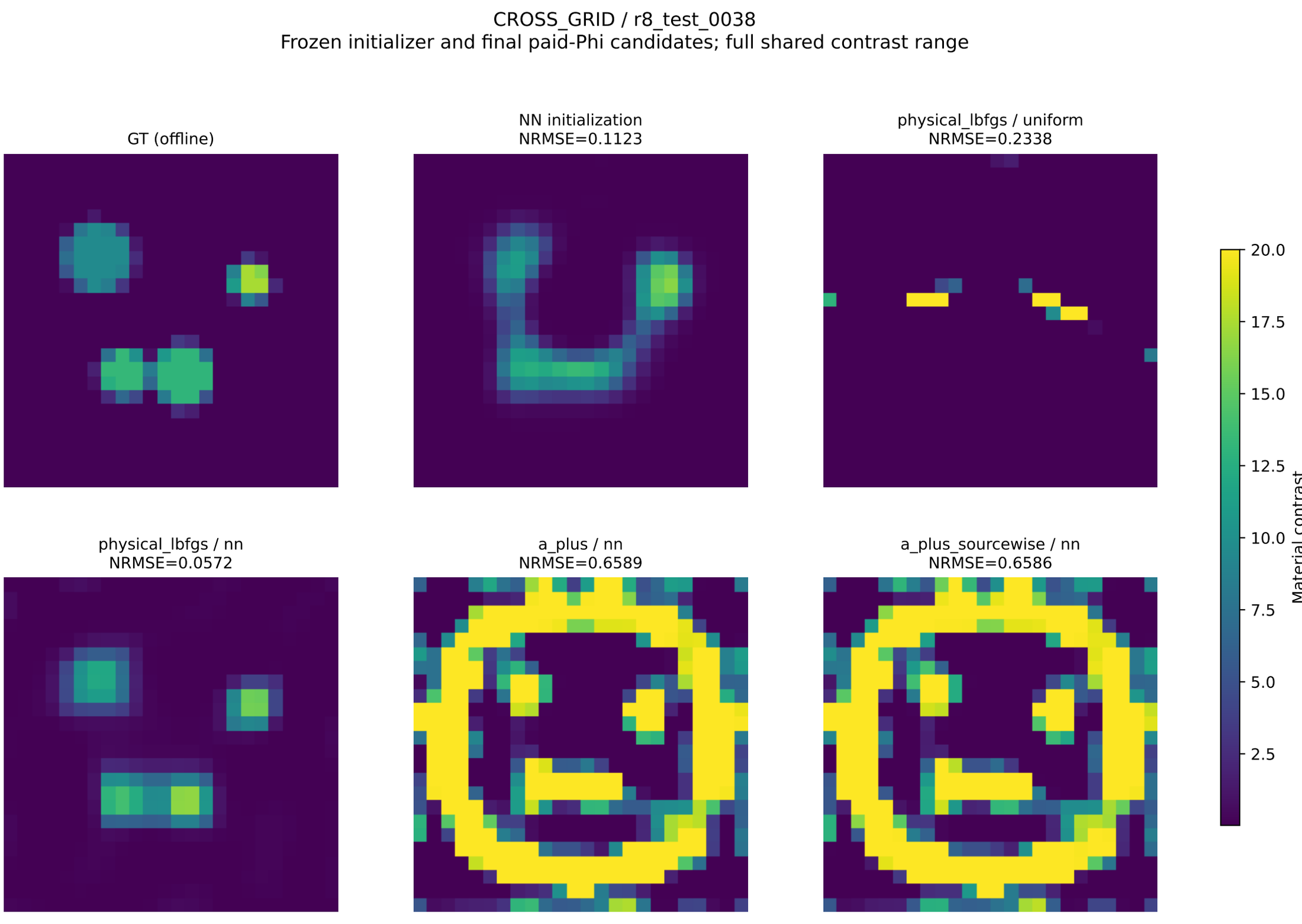

HELDOUT_FAMILY / r8_test_0033
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2746
physical_lbfgs / uniform
NRMSE=0.4478
physical_lbfgs / nn
NRMSE=0.1143
a_plus / nn
NRMSE=0.4791
a_plus_sourcewise / nn
NRMSE=0.4336
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

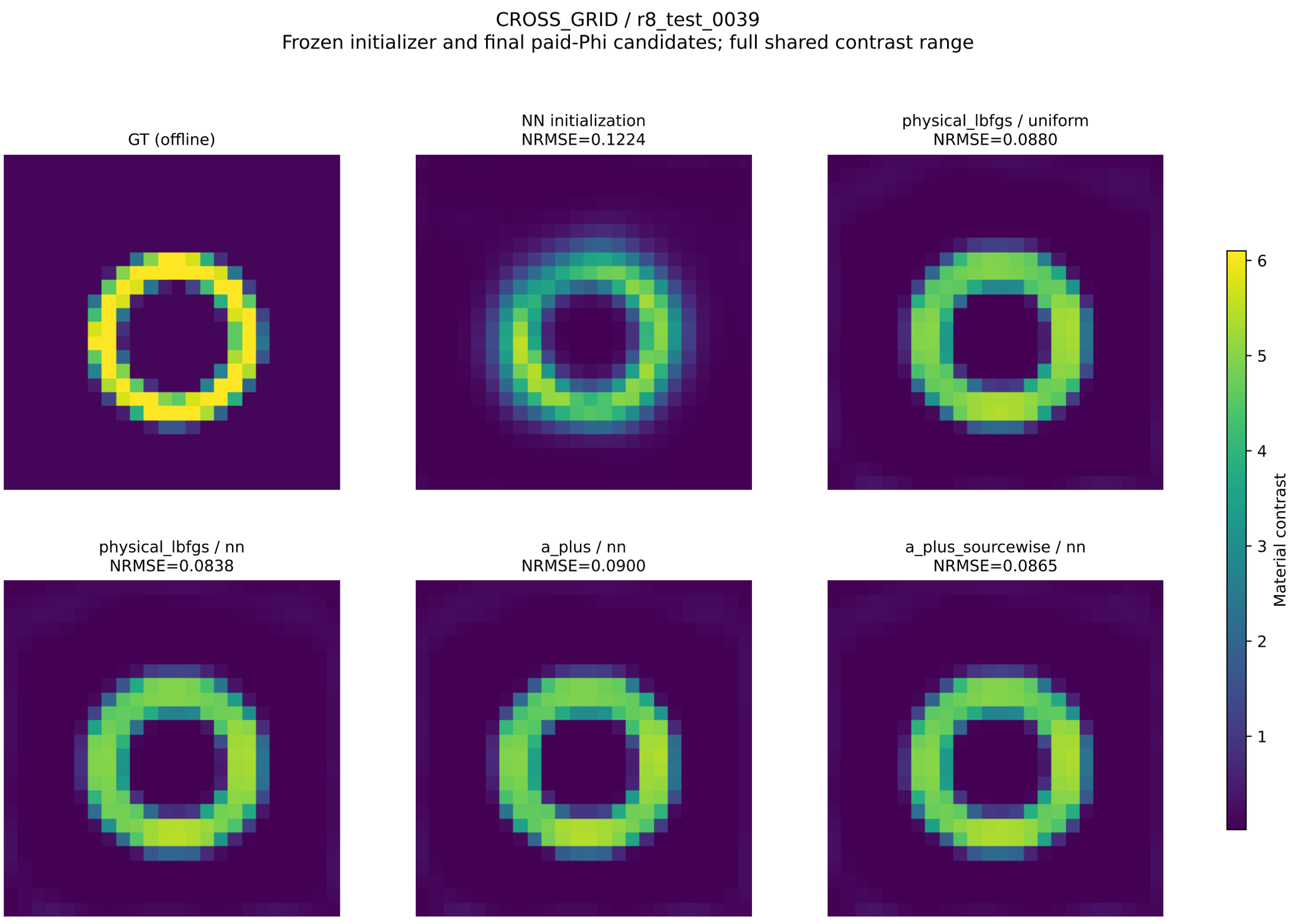

HELDOUT_FAMILY / r8_test_0034
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1808
physical_lbfgs / uniform
NRMSE=0.3540
physical_lbfgs / nn
NRMSE=0.0543
a_plus / nn
NRMSE=1.0242
a_plus_sourcewise / nn
NRMSE=1.0242
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast

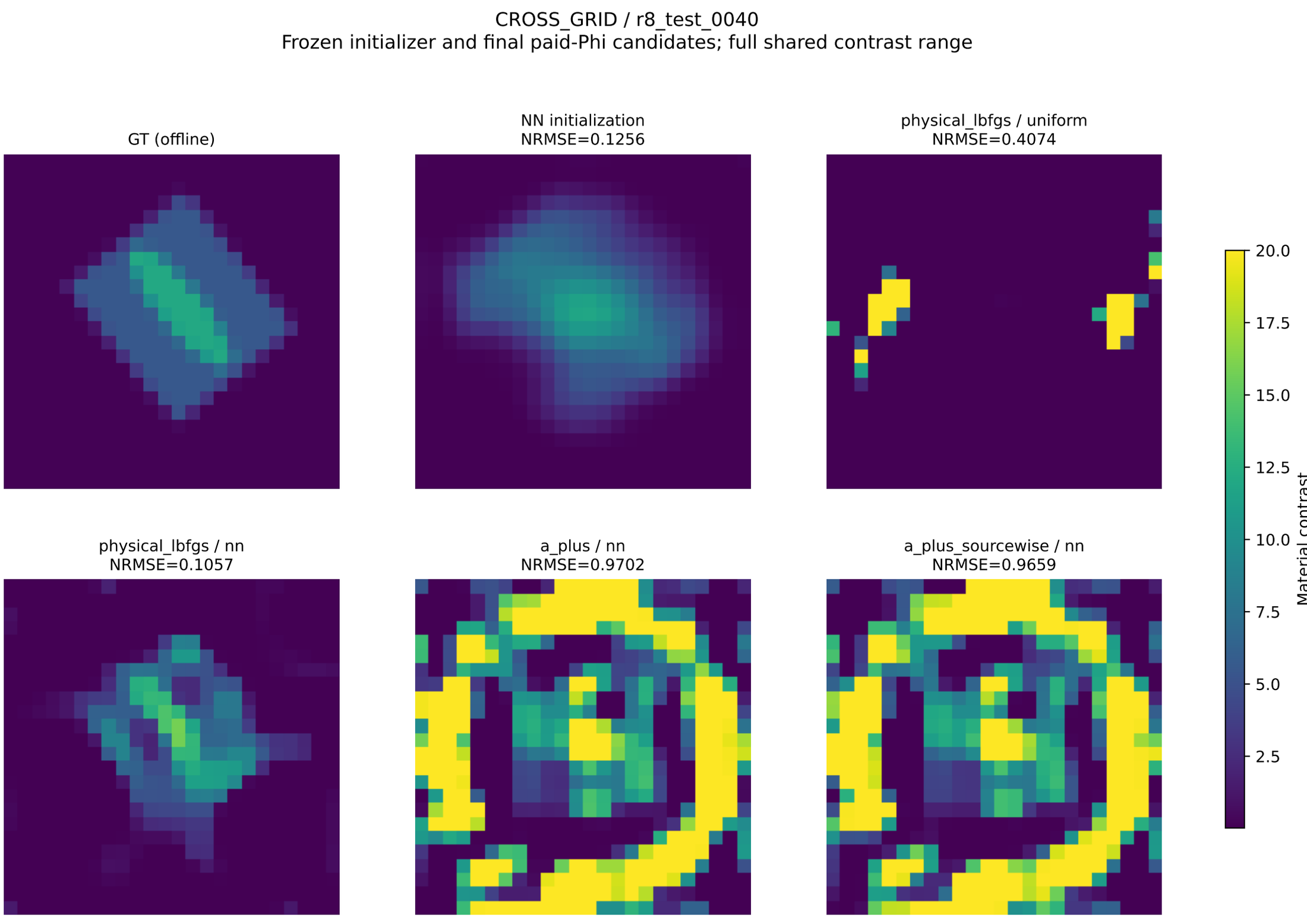

HELDOUT_FAMILY / r8_test_0035
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.2921
physical_lbfgs / uniform
NRMSE=0.5876
physical_lbfgs / nn
NRMSE=0.1705
a_plus / nn
NRMSE=1.9298
a_plus_sourcewise / nn
NRMSE=1.9302
Material contrast
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5

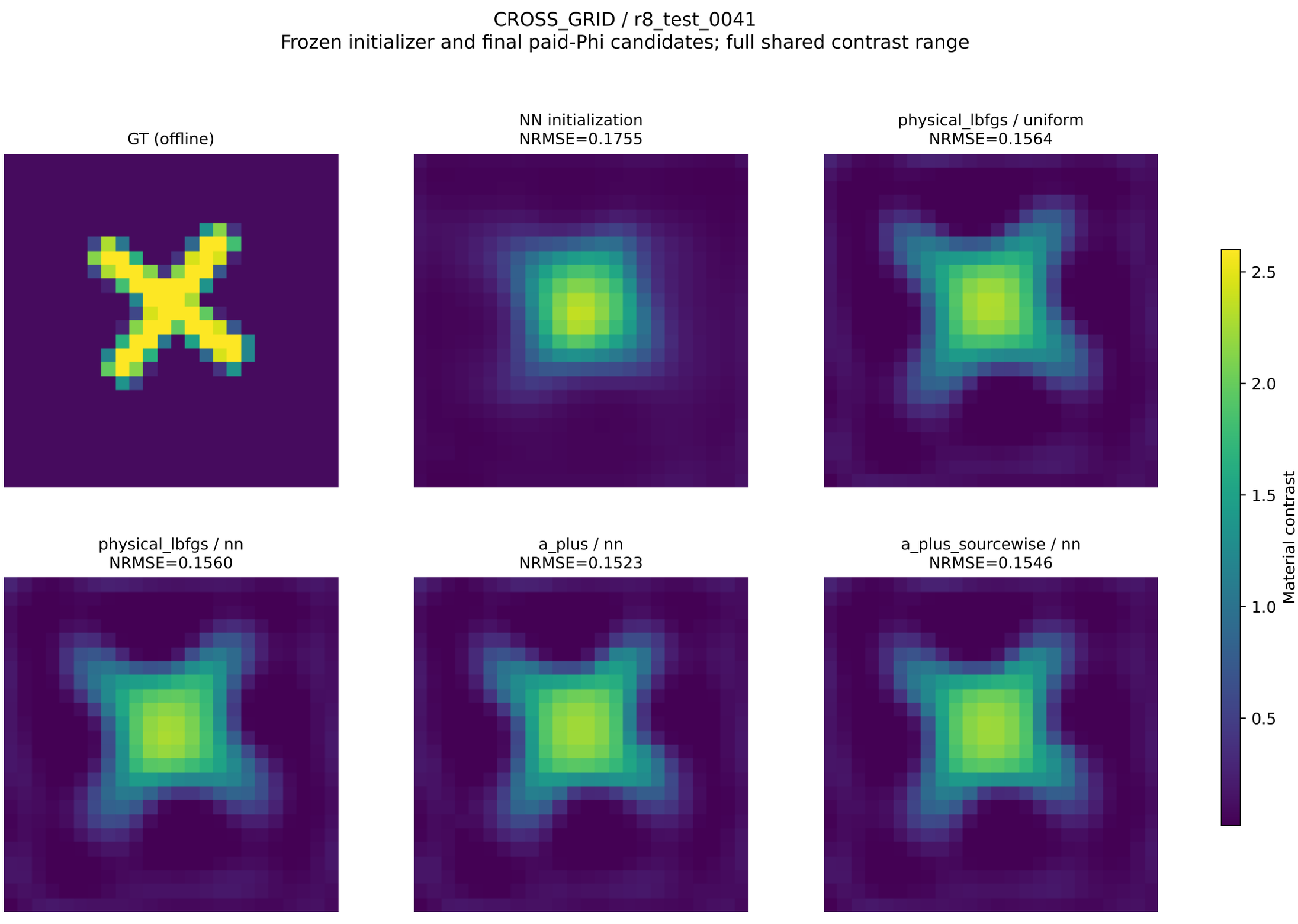

CROSS_GRID / r8_test_0036
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0899
physical_lbfgs / uniform
NRMSE=0.0512
physical_lbfgs / nn
NRMSE=0.0526
a_plus / nn
NRMSE=0.0544
a_plus_sourcewise / nn
NRMSE=0.0528
Material contrast
3.0
2.5
2.0
1.5
1.0
0.5


CROSS_GRID / r8_test_0037
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.0975
physical_lbfgs / uniform
NRMSE=0.2388
physical_lbfgs / nn
NRMSE=0.0505
a_plus / nn
NRMSE=0.0500
a_plus_sourcewise / nn
NRMSE=0.0502
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast


CROSS_GRID / r8_test_0038
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1123
physical_lbfgs / uniform
NRMSE=0.2338
physical_lbfgs / nn
NRMSE=0.0572
a_plus / nn
NRMSE=0.6589
a_plus_sourcewise / nn
NRMSE=0.6586
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast


CROSS_GRID / r8_test_0039
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1224
physical_lbfgs / uniform
NRMSE=0.0880
physical_lbfgs / nn
NRMSE=0.0838
a_plus / nn
NRMSE=0.0900
a_plus_sourcewise / nn
NRMSE=0.0865
Material contrast
1
2
3
4
5
6


CROSS_GRID / r8_test_0040
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1256
physical_lbfgs / uniform
NRMSE=0.4074
physical_lbfgs / nn
NRMSE=0.1057
a_plus / nn
NRMSE=0.9702
a_plus_sourcewise / nn
NRMSE=0.9659
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Material contrast


CROSS_GRID / r8_test_0041
Frozen initializer and final paid-Phi candidates; full shared contrast range
GT (offline)
NN initialization
NRMSE=0.1755
physical_lbfgs / uniform
NRMSE=0.1564
physical_lbfgs / nn
NRMSE=0.1560
a_plus / nn
NRMSE=0.1523
a_plus_sourcewise / nn
NRMSE=0.1546
Material contrast
2.5
2.0
1.5
1.0
0.5